



LoRa Alliance
5177 Brandin Court,
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LoRaWAN

End Device Certification

**Application Layer Packages
Test Specification**

Version 1.1.1



17 Table of Contents

18	1. Introduction	8
19	Test Condition for all application package tests	8
20	Test Notes	8
21	2. Pre-Requisite Setup Test	9
22	Test Procedure Frame Sequence Chart	9
23	3. Application Layer Clock Synchronization	11
24	PackageVersion	11
25	ForceDeviceResync and AppTimeReq	11
26	DeviceAppTimePeriodicity	12
27	Test Procedure Frame Sequence Chart	12
28	4. Remote Multicast Setup	16
29	Pre-Requisite for Remote Multicast tests	17
30	PackageVersion	17
31	McGroupSetupReq, McGroupStatusReq, McGroupDeleteReq	17
32	One multicast group setup	17
33	Two multicast group setup	18
34	Status of non-existing group	19
35	Delete non-existing group	19
36	McClassCSessionReq (for Class C devices)	19
37	With incorrect parameters	21
38	McClassBSessionReq (for Class B devices)	23
39	With incorrect parameters	25
40	Test Procedure Frame Sequence Chart	28
41	5. Fragmented Data Block Transport	50
42	PackageVersion	50
43	Fragmentation	50
44	Test Procedure Frame Sequence Chart	60
45	6. Firmware Management Protocol	74
46	PackageVersion	74
47	Device Version	74

48	Device Reboot Time	74
49	Device Reboot Countdown	75
50	Device Upgrade and Delete Image	75
51	Test Procedure Frame Sequence Chart	76
52	7. Multi-Package Access.....	80
53	Pre-Requisite for Multi Package Access protocol	81
54	PackageVersion.....	81
55	DevPackageReq.....	81
56	Multi-Package Access Protocol	81
57	Test Procedure Frame Sequence Chart	83
58		
59		
60		

61 Revision History

Version	Author	Updates	Date
1.0 Draft A	Nisha Bhaskaran	Initial Version 1.0	19 January 2022
1.0 Draft B	Nisha Bhaskaran	Incorporated Julien Catalano's comments	07 March 2022
1.0 Draft C	Nisha Bhaskaran	Added Remote Multicast Setup tests	14 March 2022
1.0 Draft D	Nisha Bhaskaran	Added Multi-Package Access, Firmware Management Protocol and Fragmented Data Block Transport tests	31 March 2022
1.0 Draft E	Nisha Bhaskaran	Addressed Julien's comments	29 October 2022
1.0 Draft F	Nisha Bhaskaran	More updates based on Julien's comments	7 November 2022
1.0.0	Nisha Bhaskaran	Reviewed and approved by DevOps- Clock Sync, Remote Multicast, Fragmentation and Firmware management	27 September 2023
1.1.0	Nisha Bhaskaran	Included Multi Package Access tests Section 7. Reviewed and approved by DevOps.	30 November 2023
1.1.1	Nisha Bhaskaran	Missing step added Section 4: Remote MulticastSetup- Step 37- added MAC command	19 September 2024



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		TxPeriodicityChangeReq to set TxPeriodicity to 50s before the Class C checks and reverted the TxPeriodicity to 5s at the end of the check.	
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64 Glossary

LoRaWAN	Protocol specification developed and maintained by the LoRa Alliance.
Certification Logo	LoRa Alliance defined logo that can be displayed on the Certified product and any documentation and marketing information about the End-Device.
LoRa Test House	Organization and corresponding facility accredited by the LoRa Alliance to perform Certification testing.
End Device	Device submitted to a LoRa-Alliance Authorized Test House (ATH) for Certification.

65 Abbreviations & Acronyms

ABP	Activation by Personalization
ADR	Adaptive Data Rate
AS	Application Server
ATH	Authorized Test House
AWG	Arbitrary Waveform Generator
DR	Data Rate
DUT	Device Under Test
ERP	Equivalent Radiated Power compared to a dipole antenna (expressed in dBd)
EIRP	Equivalent Isotropic Radiated Power: $ERP = EIRP - 2.15\text{dB}$ (expressed in dBi)
ETSI	European Telecommunications Standards Institute
FSK	Frequency Shift Keying modulation technique.
FUOTA	Firmware Update Over The Air



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GW	Gateway
LCTT	LoRaWAN Certification Test Tool
MAC	Media Access Control
NS	Network Server
OTAA	Over-the-Air Activation
TCL	Test Control Layer
TRP	Total Radiated Power

66 Definitions of terms used in this document

XXXX	X is a valid number which would vary based on the region being tested
DC	Dynamic Channel
FC	Fixed Channel
MinDR	Minimum Data Rate for the region tested, as specified in the Regional Parameters Specification RP2 1.0.4
Max125kHzDR	Maximum Data Rate using 125 kHz bandwidth for the region tested, as specified in the Regional Parameters Specification RP2 1.0.4
R	Repeat until the condition specified within brackets [] is completed

67 References

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- [1] LoRaWAN L2 Link Layer Specification TS001 v1.0.4.
- [2] LoRaWAN Regional Parameters Specification RP002 v1.0.4.
- [3] LoRaWAN Certification Protocol Specification TS009 v1.2.0 draftF.
- [4] LoRaWAN End Device Certification Requirements for All Regions v1.6
- [5] LoRaWAN Remote Multicast Setup Specification TS005-2.0.0
- [6] LoRaWAN Application Layer Clock Synchronization TS003-v2.0.0

69

70

71



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72 [7] TS007-Multi_Package_Access-v1.0.0
73 [8] TS006-Firmware_Management_Protocol-v1.0.0
74 [9] TS004-Fragmented_Data_Block_Transport_v2.0.0
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1. Introduction

This document specifies the testing requirements for an End-Device for the Firmware Update Over the Air Specifications. This document describes the test cases for all the following specifications-

1. TS003-Application_Layer_Clock_Synchronization_v2.0.0
2. TS004-Fragmented_Data_Block_Transport_v2.0.0
3. TS005-Remote_Multicast_Setup_v2.0.0
4. TS006-Firmware_Management_Protocol-v1.0.0
5. TS007-Multi_Package_Access-v1.0.0

The Certification for each of these specifications listed above can be performed either individually (for the first 4 specifications mentioned above) or collectively (for TS007) depending on the End-Device's supported features.

Test Condition for all application package tests

The TCL must handle the LoRaWAN protocol like a real network server, unless specified otherwise in the tests.

Throughout the testing, the DUT MAY send the MAC commands and uplink frames and the TCL must respond appropriately as specified below:

- If the DUT sends a Confirmed frame anytime, the TCL must respond with an Acknowledgement.
- If the DUT sends a DeviceTimeReq MAC command or any other DUT triggered MAC command anytime, the TCL must respond with a DeviceTimeAns MAC answer or the appropriate MAC answer immediately.

Test Notes

1. **Default Data Rate:** All LinkADRReq MAC commands in this document must set the Data Rate to Max125kHzDR, unless specified otherwise.
2. **MIC check:** When the tool encounters an invalid MIC, it must fail the specific test being performed.
3. **Missed MAC or Certification Protocol commands:** During testing, if MAC or Certification Protocol commands are missed by the DUT, the command must be re-sent by the TCL up to two additional times before failing the test.

4. **Missed RxAppCntReq command:** When an RxAppCntReq command is sent from the TCL to the DUT, if the response is not received from the DUT, the request must be re-sent up to two additional times.
5. **Downlink after sending MAC commands which require a response:** Unless specified otherwise, for all MAC response commands which require a response from the TCL (namely, RXParamSetupAns, RXTimingSetupAns, DLChannelAns), the TCL must send a downlink immediately after receiving this MAC response command.
6. **Uplink response timing:** The DUT responses to the TCL downlinks may not necessarily happen at the next uplink from the DUT. The TCL needs to wait for a DUT uplink from the appropriate FPort for the application package being tested. If the DUT sends 3 uplinks for a different FPort, the TCL must resend the downlink and wait up to 3 more uplinks for a response. The downlinks are repeated up to 2 additional times.

2. Pre-Requisite Setup Test

The Pre-requisite setup tests need to be run as a first step before performing any of the other test cases mentioned in this document, to speed up the test execution.

The DUT is brought to the appropriate state to conduct the FUOTA tests by executing these test cases.

The Pre-requisite tests include-

1. Change TXPeriodicity to 5 seconds
2. Switch to Unconfirmed uplink frames (if not already)
3. Turn on the ADR bit (if not already set)
4. Change the uplink DataRate to Max125KHz

Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Confirmed or Unconfirmed frame	→		
	The TCL sends Unconfirmed frame in RX1 window	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 5 Payload = [0x]0601 If uplink was a Confirmed frame, send ACK	Uplink Periodicity set to 5s
2	DUT sends Confirmed or Unconfirmed frame	→		
	If DUT sent a Confirmed frame, then The TCL sends Unconfirmed frame Else, this step must be skipped	←	CP-CMD TxFramesCtrlReq FPort = 224 Frame type = Unconfirmed Payload = [0x]0701 If DUT sent Confirmed uplink frame, TCL must Acknowledge	Switch to Unconfirmed frames
3	DUT sends Unconfirmed frame	→		
	If FCtrl ADR Bit = false, then The TCL sends Unconfirmed frame Else, this step is skipped	←	CP-CMD AdrBitChangeReq-ON FPort = 224 Payload = [0x]0401	
4	DUT sends Unconfirmed frame	→	FCtrl ADR bit = true	Turn on the ADR bit
	The TCL sends Unconfirmed frame	←	CMD LinkADRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXX ChMaskCntl: DC = 0, FC = 6 ChMask: DC - Enable only default channels FC = [0x]00FF	For FC device using 8-channel gateway, refer to Section 2 of [4] for ChMask settings
5	DUT sends Unconfirmed frame	→	CMD LinkADRAns Payload = [0x]0307	Changed the DR to Max125kHz DR

3. Application Layer Clock Synchronization

Test Purpose: This test verifies all the Clock synchronization commands from TS003-Application_Layer_Clock_Synchronization_v2.0.0 Specification [6].

Note: It is recommended that a default FPort value of 202 be used to promote interoperability. The default FPort value is obtained from the Certification Questionnaire.

PackageVersion

- TCL sends the command PackageVersionReq to the DUT
- DUT must respond with the command PackageVersionAns with Package Identifier = 1 and Package Version = 2

ForceDeviceResync and AppTimeReq

Note: In these tests, the TCL shall use the DeviceTime value sent in AppTimeReq to check that the DUT has synchronized its clock correctly

- In order to synchronize the clock, TCL sends the ForceDeviceResyncCmd command to the DUT.
- DUT sends AppTimeReq until the clock is synchronized – difference of time is -1s to +1s.
- TCL sends the ForceDeviceResyncCmd command again to the DUT to confirm that the clock is in sync.
- DUT sends AppTimeReq and TCL responds with the AppTimeAns.
- TCL sends the ForceDeviceResyncCmd command to the DUT with NbTransmissions = 3
- DUT must send 3 AppTimeReq commands with AnsRequired = 0
- TCL sends the ForceDeviceResyncCmd command to the DUT with NbTransmissions = 0
- DUT must silently discard the ForceDeviceResyncCmd command
- TCL sends the ForceDeviceResyncCmd command to the DUT with NbTransmissions = 3
- DUT must send the AppTimeReq command with AnsRequired = 0
- TCL immediately responds with the AppTimeAns command
- DUT must stop sending AppTimeReq commands.
- DUT must send the AppTimeReq command.
- TCL responds with an AppTimeAns command with the TokenAns which is incorrect
- DUT must ignore the command and continue sending uplinks
- DUT must send the AppTimeReq command.
- TCL responds with an AppTimeAns command with TimeCorrection overflow (0x7FFFFFFF)
- DUT must send a new AppTimeReq command.
- TCL sends the remaining TimeCorrection on another AppTimeAns command.
- DUT must send the AppTimeReq command.
- TCL responds with an AppTimeAns command with TimeCorrection overflow (0x80000000)

- DUT must send a new AppTimeReq command.
- TCL sends the remaining TimeCorrection on another AppTimeAns command.

Overflow of TokenReq test

- TCL sends the ForceDeviceResyncCmd command to the DUT
- DUT must send the AppTimeReq command
- TCL responds with AppTimeAns
- Repeat the above steps for a max of 16 AppTimeReq requests until TokenReq overflows. Once the TokenReq overflows, the TokenReq counter must be reset to 0.

DeviceAppTimePeriodicity

- TCL sends the command DeviceAppTimePeriodicityReq to the DUT with Periodicity set to 128 seconds (i.e. Period = 0)
- DUT must respond with DeviceAppTimePeriodicityAns message with DeviceTime.
- If NotSupported bit = 0, DUT must send at least 1 Unconfirmed frame containing AppTimeReq after TCLTimestamp +128-30 and before TCLTimestamp +128 +30 seconds. Note: The DUT may send AppTimeReq commands at any other time as well.
- If the NotSupported bit =1, test ends here.
- If the AnsRequired bit = 1, TCL responds with an AppTimeAns command and then waits 128 seconds +/- 30 seconds. If the AnsRequired = 0, TCL responds with the AppTimeAns only if the DUT indicated current clock timing drifts above a certain application specific threshold. If the TCL sends AppTimeAns, the TokenAns must match the TokenReq value of the AppTimeReq command which is being answered.
- DUT sends at least 1 Unconfirmed frame containing AppTimeReq after TCLTimestamp +128-30 and before TCLTimestamp +128 +30+TimeOnAir(3) seconds with its TokenReq value incremented. Note: The DUT may send AppTimeReq commands at any other time as well.
- If the AnsRequired bit = 1, TCL responds with an AppTimeAns command and then waits 128 seconds +/- 30 seconds. If the AnsRequired = 0, TCL responds with the AppTimeAns only if the DUT indicated current clock timing drifts above a certain application specific threshold. If the TCL sends AppTimeAns, the TokenAns must match the TokenReq value of the AppTimeReq command which is being answered.

Test Procedure Frame Sequence Chart

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD PackageVersionReq Payload = [0x]00	PackageVersionReq test
2	DUT sends Unconfirmed frame	→	CMD PackageVersionAns Payload = [0x]000102	Package Identifier = 1 Package Version = 2
	The TCL sends Unconfirmed frame	←	CMD ForceDeviceResyncCmd NbTransmissions = 1 Payload = [0x]XX	
3	DUT sends Unconfirmed frame	R [until clock is sync'ed – difference of time is -1s to +1s]	CMD AppTimeReq AnsRequired = 0 (Mandatory for the first AppTimeReq directly after ForceDeviceResyncCmd) Payload = [0x]01XXXXXXXXXX	
	The TCL sends Unconfirmed frame		CMD AppTimeAns Payload = [0x]01XXXXXXXXXX	
4	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD ForceDeviceResyncCmd NbTransmissions = 1 Payload = [0x]XX	
5	DUT sends Unconfirmed frame	→	CMD AppTimeReq AnsRequired = 0 Payload = [0x]01XXXXXXXXXX	
6	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD ForceDeviceResyncCmd NbTransmissions = 3 Payload = [0x]XX	
7	DUT sends Unconfirmed frame This uplink must be repeated 3 times	→ R[3]	CMD AppTimeReq AnsRequired = 0 Payload = [0x]01XXXXXXXXXX TokenReq must be incremented first time. The TokenReq must not be incremented in the repetitions.	
8	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD ForceDeviceResyncCmd NbTransmissions = 0 Payload = [0x]XX	
9	DUT sends Unconfirmed frame	→		ForceDeviceResyncCmd must be silently discarded
	The TCL sends Unconfirmed frame	←	CMD ForceDeviceResyncCmd NbTransmissions = 3 Payload = [0x]XX	

10	DUT sends Unconfirmed frame	→	CMD AppTimeReq AnsRequired = 0 Payload = [0x]01XXXXXXXXXX TokenReq must not be incremented	
	The TCL sends Unconfirmed frame	←	CMD AppTimeAns Payload = [0x]01XXXXXXXXXX	
11	DUT sends Unconfirmed frame	→		AppTimeReq transmission stops
	The TCL sends Unconfirmed frame	←	CMD ForceDeviceResyncCmd NbTransmissions = 1 Payload = [0x]XX	
12	DUT sends Unconfirmed frame	→	CMD AppTimeReq AnsRequired = 0 Payload = [0x]01XXXXXXXXXX TokenReq must be incremented	
	TCL sends Unconfirmed frame	←	CMD AppTimeAns TokenAns must be incorrect Payload = [0x]01XXXXXXXXXX	
13	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD ForceDeviceResyncCmd NbTransmissions = 1 Payload = [0x]XX	
14	DUT sends Unconfirmed frame	→	CMD AppTimeReq AnsRequired = 0 Payload = [0x]01XXXXXXXXXX TokenReq must NOT be incremented due to the incorrect TokenAns	
	TCL sends Unconfirmed frame	←	CMD AppTimeAns TimeCorrection overflow (0x7FFFFFFF) Payload = [0x]01XXXXXXXXXX	
15	DUT sends Unconfirmed frame	→	CMD AppTimeReq Payload = [0x]01XXXXXXXXXX TokenReq must be incremented	DUT must send a new AppTimeReq command
	TCL sends Unconfirmed frame	←	CMD AppTimeAns TimeCorrection = remaining TimeCorrection value Payload = [0x]01XXXXXXXXXX	
16	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD ForceDeviceResyncCmd NbTransmissions = 1 Payload = [0x]XX	

17	DUT sends Unconfirmed frame	→	CMD AppTimeReq AnsRequired = 0 Payload = [0x]01XXXXXXXXXX TokenReq must be incremented	
	TCL sends Unconfirmed frame	←	CMD AppTimeAns TimeCorrection overflow (0x8000000) Payload = [0x]01XXXXXXXXXX	
18	DUT sends Unconfirmed frame	→	CMD AppTimeReq Payload = [0x]01XXXXXXXXXX TokenReq must be incremented	DUT must send a new AppTimeReq command
	TCL sends Unconfirmed frame	←	CMD AppTimeAns TimeCorrection = remaining TimeCorrection value Payload = [0x]01XXXXXXXXXX	
19	DUT sends Unconfirmed frame	R [for a max of 16 AppTimeReq until TokenReq overflows]		
	The TCL sends Unconfirmed frame		CMD ForceDeviceResyncCmd NbTransmissions = 1 Payload = [0x]XX	
20	DUT sends Unconfirmed frame		CMD AppTimeReq AnsRequired = 0 Payload = [0x]01XXXXXXXXXX TokenReq must be incremented	
	TCL sends Unconfirmed frame Repeat the previous step and this step for a max of 16 AppTimeReq requests (including the other AppTimeReq requests sent in this test above), until TokenReq overflows.		CMD AppTimeAns Payload = [0x]01XXXXXXXXXX When TokenReq overflows, the TokenReq field counter must be reset to 0	
21	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD DeviceAppTimePeriodicityReq Period = 0 (i.e. Periodicity = 128sec) Payload = [0x]02XX	DeviceAppTimePeriodicityReq test
22	DUT sends Unconfirmed frame	→	CMD DeviceAppTimePeriodicityAns Payload = [0x]02XXXXXXXXXX	TCL starts the time stamp for checking the 128s periodicity

23	If NotSupported bit = 0, DUT sends at least 1 Unconfirmed frame containing AppTimeReq after TCLTimestamp +128-30 and before TCLTimestamp +128 +30+3* seconds * <i>TimeOnAir</i> If NotSupported bit = 1, test ends here Note: Other uplinks may be sent by the DUT at any time while waiting for the condition to be fulfilled.	→	CMD AppTimeReq Payload = [0x]01XXXXXXXXXX	AppTimeReq test TCL checks the AppTimeReq periodicity = 128 +/-30 seconds. TCL restarts the time stamp check.
	If AnsRequired = 1, TCL sends Unconfirmed frame If AnsRequired = 0, TCL sends Unconfirmed frame only if the DUT indicated current clock timing drifts above a certain application specific threshold	←	CMD AppTimeAns Payload = [0x]01XXXXXXXXXX TokenAns must match the TokenReq	
24	DUT sends at least 1 Unconfirmed frame containing AppTimeReq after TCLTimestamp +128-30 and before TCLTimestamp +128 +30+3* seconds. * <i>TimeOnAir</i> Repeat this step twice to verify that the AppTimeReq commands are sent periodically. Note: Other uplinks and AppTimeReq may be sent by the DUT at any time while waiting for the condition to be fulfilled.	→ R[2]	CMD AppTimeReq Payload = [0x]01XXXXXXXXXX TokenReq must be incremented (modulo 16) if AppTimeAns is sent by the TCL in the previous step	AppTimeReq resent periodically TCL checks the AppTimeReq periodicity = 128 +/-30 seconds.
	If AnsRequired = 1, TCL sends Unconfirmed frame If AnsRequired = 0, TCL sends Unconfirmed frame only if the DUT indicated current clock timing drifts above a certain application specific threshold	← R[2]	CMD AppTimeAns Payload = [0x]01XXXXXXXXXX TokenAns must match the TokenReq	
25	DUT sends Unconfirmed frame	→		

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210 4. Remote Multicast Setup

211 Test Purpose: This test verifies all the Remote Multicast Setup commands from TS005-
212 Remote_Multicast_Setup_2.0.0 Specification [5].

213

214 **Note:** It is recommended that a default FPort value of 200 be used to promote interoperability. The
215 default FPort value is obtained from the Certification Questionnaire.

Pre-Requisite for Remote Multicast tests

DUT clock synchronization:

- TCL sends CP-CMD DeviceTimeReq on FPort 224 to DUT.
- DUT must issue a DeviceTimeReq command.

PackageVersion

- TCL sends the command PackageVersionReq to the DUT
- DUT must respond with the command PackageVersionAns with Package Identifier = 2 and Package Version = 2

McGroupSetupReq, McGroupStatusReq, McGroupDeleteReq

One multicast group setup

- TCL creates a multicast session on the DUT using McGroupSetupReq with the following parameters:
 - McGroupID = 0
 - McAddr = 0x00ABCDEF
 - McKey = 0x0102030405060708090A0B0C0D0E0F10
 - minMcFCount = 0
 - maxMcFCount = 0x0000FFFF
- DUT must respond with the McGroupSetupAns command without any errors, with the correct McGroupID.
- TCL sends the McGroupStatusReq command to the DUT, with ReqGroupMask = 0x1, requesting the first multicast group status.
- The DUT must respond with the McGroupStatusAns command, with the status field mentioning a unique group defined on the DUT, included in the answer. The correct McGroupID and McAddr field (single record) may be sent following the status byte.
- TCL sends the McGroupDeleteReq command to the DUT to request the deletion of the earlier defined group.
- The DUT must respond with McGroupDeleteAns command, mentioning no error and the correct McGroupID.
- TCL sends the McGroupStatusReq command to the DUT, with ReqGroupMask = 0x1, requesting the first multicast group status.
- The DUT must respond with the McGroupStatusAns command confirming the deletion of the group.

Two multicast group setup

- TCL creates a multicast session on the DUT using McGroupSetupReq with the following parameters:
 - McGroupID = 0
 - McAddr = 0x00ABCDEF
 - McKey = 0x0102030405060708090A0B0C0D0E0F10
 - minMcFCount = 0
 - maxMcFCount = 0x0000FFFF
- DUT must respond with the McGroupSetupAns command without any errors, with the correct McGroupID.
- TCL creates a second multicast session on the DUT using McGroupSetupReq with the following parameters:
 - McGroupID = 1
 - McAddr = 0x00123456
 - McKey = 0x100F0E0D0C0B0A090807060504030201
 - minMcFCount = 0
 - maxMcFCount = 0x0000FFFF
- DUT must respond with the McGroupSetupAns command.
 - If the DUT supports multiple multicast group, it must answer without any errors, and with the correct McGroupID. The test can continue.
 - If the DUT does not support multiple multicast group (check the Questionnaire/LWCS), it must answer with IDerror = 1, with the given McGroupID. Test ends here. Skip to the tests in the next section (Status of non-existing group).
- TCL sends the McGroupStatusReq command to the DUT, with ReqGroupMask = 0x3, requesting the two first multicast groups status.
- The DUT must respond with the McGroupStatusAns command, with the status field mentioning the two groups defined on the DUT, included in the answer. The correct McGroupID and McAddr field (two records) may be sent following the status byte.

To test truncation

- TCL sends a LinkADRReq to change the DataRate to the lowest DR. DUT must send a successful LinkADRAAns.
- TCL sends the McGroupStatusReq command to the DUT, with ReqGroupMask = 0x3, requesting the two first multicast groups status.
- The DUT must respond with the McGroupStatusAns command, with the status field mentioning only the groups which fit in the answer. The DUT must discard the last groups that do not fit in the answer. The AnsMaskGroup must be sent accordingly. The correct McGroupID and McAddr field of the records which fit must be sent following the status byte.

- 294 • TCL sends a LinkADRRReq to change the DataRate to the Max125kHzDR. DUT must
295 send a successful LinkADRAns.
296
297 • TCL sends the McGroupDeleteReq command to the DUT to request the deletion of
298 the first defined group using McGroupID = 1.
299 • The DUT must respond with McGroupDeleteAns command, mentioning no error and
300 the correct McGroupID.
301 • TCL sends the McGroupStatusReq command again to the DUT, with ReqGroupMask
302 = 0x3, requesting the two first multicast groups status.
303 • The DUT must respond with the McGroupStatusAns command, with the status field
304 mentioning only the second group defined on the DUT, included in the answer. The
305 correct McGroupID and McAddr field (single record) may be sent following the
306 status byte.

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309 Status of non-existing group

- 310 • TCL sends the McGroupDeleteReq command to the DUT to request the deletion of
311 the first defined group using McGroupID = 0.
312 • The DUT must respond with McGroupDeleteAns command, mentioning no error and
313 the correct McGroupID.
314 • TCL sends the McGroupStatusReq command to the DUT, with ReqGroupMask = 0x8,
315 requesting the fourth multicast group status.
316 • The DUT must respond with the McGroupStatusAns command, with the status field
317 mentioning the total number of groups defined on the DUT, and no group included
318 in the answer (AnsGroupMask = 0). No record must be sent following the status
319 byte, answer size must be 1 byte.

320

321 Delete non-existing group

- 322 • TCL sends the McGroupDeleteReq command to the DUT to request the deletion of
323 the non-existing group using McGroupID = 1.
324 • The DUT must respond with McGroupDeleteAns command, mentioning that the
325 multicast group McGroupID = 1 is undefined (McGroupUndefined = 1).

326

327 McClassCSessionReq (for Class C devices)

328 Note: The tests in this section must be run only if the DUT supports Class C

329

- 330 ○ TCL creates a multicast session on the DUT using McGroupSetupReq with the
331 following parameters:

- 332 ▪ McGroupID = 0
- 333 ▪ McAddr = 0x00ABCDEF
- 334 ▪ McKey = 0x0102030405060708090A0B0C0D0E0F10
- 335 ▪ minMcFCount = 0
- 336 ▪ maxMcFCount = 0x0000FFFF
- 337 ○ DUT must respond with the McGroupSetupAns command without any errors, with
- 338 the correct McGroupID.
- 339 ○ TCL sends a McGroupStatusReq command on RX2 window.
- 340 ○ The DUT must respond correctly with the McGroupStatusAns command.
- 341 ○ TCL sends the McClassCSessionReq command to the DUT to setup a temporary Class
- 342 C multicast session with
- 343 ▪ McGroupID = 0
- 344 ▪ SessionTime = 30 seconds from the current time
- 345 ▪ SessionTimeOut = 32 seconds
- 346 ▪ DLFreq = Any allowed frequency other than the default RX2 frequency (refer
- 347 to Regional Parameters Specification)
- 348 ▪ DR = Max125KHzDR or Max500kHzDR (refer to Regional Parameters
- 349 Specification)
- 350 ○ The DUT must respond with McClassCSessionAns command with
- 351 ▪ McGroupID = 0
- 352 ▪ DRError = 0
- 353 ▪ FreqError = 0
- 354 ▪ McGroupUndefined = 0
- 355 ▪ StartMissed = 0
- 356 ▪ Reserved for future errors = 0
- 357 ▪ TimeToStart = the number of seconds from the McClassCSessionAns uplink to
- 358 the beginning of the multicast session
- 359 ○ TCL sends CP-CMD RxAppCntReq to DUT
- 360 ○ DUT must respond with RxAppCntAns
- 361 ○ After the SessionTime, TCL sends repeated CP-CMD TxFramesCtrlReq with
- 362 FrameType=0 (No change) using multicast address and session context.
- 363 ○ Wait until SessionTimeOut.
- 364 ○ TCL sends CP-CMD RxAppCntReq to DUT.
- 365 ○ DUT must respond with RxAppCntAns with the RxAppCnt field incremented by the
- 366 number of downlinks sent by the TCL. (Tolerance of 5%)
- 367
- 368 ○ The TCL sends 3 Unconfirmed frames using multicast and session context during the
- 369 RXC window after the multicast session has closed.
- 370 ○ DUT must ignore the downlinks.
- 371 ○ TCL deletes the McGroupID = 0 and creates a new group with McGroupID = 0 again,
- 372 with minMcFCnt = 5 and maxMcFCnt = 20

- 373 ○ TCL creates a new McClassCSession
- 374 ○ After the Session time starts, TCL sends Unconfirmed frame using multicast and
- 375 session context with McFCnt < minMcFCnt.
- 376 ○ DUT must ignore this downlink.
- 377 ○ TCL sends Unconfirmed frame using multicast and session context with McFCnt >
- 378 maxMcFCnt.
- 379 ○ DUT must ignore this downlink.
- 380 ○ TCL sends Confirmed frame using multicast and session context with McFCnt >
- 381 minMcFCnt and McFCnt > maxMcFCnt.
- 382 ○ DUT must ignore this downlink.
- 383 ○ TCL sends Unconfirmed MAC-CMD frame using multicast and session context.
- 384 ○ DUT must ignore this downlink.
- 385 ○ TCL sends Unconfirmed McGroupSetupReq frame using multicast and session
- 386 context.
- 387 ○ DUT must ignore this downlink.

388 **With multiple groups**

- 390 • TCL sends McGroupSetupReq to create 2 McGroups ID = 0 and 1.
- 391 • DUT responds with McGroupSetupAns successfully.
- 392 • TCL sends McClassCSessionReq to create 2 sessions for each McGroupID 0 and 1 with
- 393 the same parameters.
- 394 • DUT responds with McClassCSessionAns successfully.
- 395 • TCL changes the Periodicity of uplinks to 240 seconds.
- 396 • Wait for SessionTime.
- 397 • TCL sends repeated downlinks alternatively on McGroupID 0 and 1.
- 398 • TCL reverts the Periodicity of uplinks to 5 seconds.
- 399 • TCL checks the counter using CP-CMD RxAppCntAns to confirm that the downlinks are
- 400 received from both the multicast groups with a tolerance of 5%.
- 401 • TCL deletes McGroupID = 1 using McGroupDeleteReq command.
- 402 • DUT responds with McGroupDeleteAns command.

403 **With incorrect parameters**

- 404 • TCL sends the McClassCSessionReq command to the DUT to setup a temporary Class C
- 405 multicast session with
 - 406 ○ McGroupID = 1 (invalid)
 - 407 ○ SessionTime = 8 seconds from the current time
 - 408 ○ SessionTimeOut = 32 seconds
 - 409 ○ DLFreq = Any allowed frequency other than the default RX2 frequency (refer to
 - 410 Regional Parameters Specification)
 - 411 ○ DR = Max125KHzDR or Max500kHzDR (refer to Regional Parameters Specification)
- 412 • The DUT must respond with McClassCSessionAns command with

- 413 ○ McGroupID = 1
- 414 ○ DRError = 0
- 415 ○ FreqError = 0
- 416 ○ McGroupUndefined = 1
- 417 ○ StartMissed = 0
- 418 ○ Reserved for future errors = 0
- 419 • TCL sends the McClassCSessionReq command to the DUT to setup a temporary Class C
- 420 multicast session with
- 421 ○ McGroupID = 0
- 422 ○ SessionTime = Before the current time (invalid)
- 423 ○ SessionTimeout = 32 seconds
- 424 ○ DLFreq = Any allowed frequency other than the default RX2 frequency (refer to
- 425 Regional Parameters Specification)
- 426 ○ DR = Max125KHzDR or Max500kHzDR (refer to Regional Parameters Specification)
- 427 • The DUT must respond with McClassCSessionAns command with
- 428 ○ McGroupID = 0
- 429 ○ DRError = 0
- 430 ○ FreqError = 0
- 431 ○ McGroupUndefined = 0
- 432 ○ StartMissed = 1
- 433 ○ Reserved for future errors = 0
- 434 • TCL sends the McClassCSessionReq command to the DUT to setup a temporary Class C
- 435 multicast session with
- 436 ○ McGroupID = 0
- 437 ○ SessionTime = 8 seconds from the current time
- 438 ○ SessionTimeout = 32 seconds
- 439 ○ DLFreq = Invalid frequency (invalid)
- 440 ○ DR = Max125KHzDR or Max500kHzDR (refer to Regional Parameters Specification)
- 441 • The DUT must respond with McClassCSessionAns command with
- 442 ○ McGroupID = 0
- 443 ○ DRError = 0
- 444 ○ FreqError = 1
- 445 ○ McGroupUndefined = 0
- 446 ○ StartMissed = 0
- 447 ○ Reserved for future errors = 0
- 448 • TCL sends the McClassCSessionReq command to the DUT to setup a temporary Class C
- 449 multicast session with
- 450 ○ McGroupID = 0
- 451 ○ SessionTime = 8 seconds from the current time
- 452 ○ SessionTimeout = 32 seconds

- DLFreq = Any allowed frequency other than the default RX2 frequency (refer to Regional Parameters Specification)
- DR = Invalid DR
- The DUT must respond with McClassCSessionAns command with
 - McGroupID = 0
 - DRError = 1
 - FreqError = 0
 - McGroupUndefined = 0
 - StartMissed = 0
 - Reserved for future errors = 0
- TCL sends the McClassCSessionReq command to the DUT to setup a temporary Class C multicast session with
 - McGroupID = 0
 - SessionTime = Current time + 16777215 + 20 seconds
 - SessionTimeout = 32 seconds
 - DLFreq = Any allowed frequency other than the default RX2 frequency (refer to Regional Parameters Specification)
 - DR = Max125KHzDR or Max500kHzDR (refer to Regional Parameters Specification)
- The DUT must respond with McClassCSessionAns command with
 - McGroupID = 0
 - DRError = 0
 - FreqError = 0
 - McGroupUndefined = 0
 - StartMissed = 0
 - Reserved for future errors = 0
 - TimeToStart = 0xFFFFFFFF
- TCL sends the McGroupDeleteReq command to the DUT to request the deletion of the group created using McGroupID = 0.
- The DUT must respond with McGroupDeleteAns command, mentioning no error and the correct McGroupID.

McClassBSessionReq (for Class B devices)

Note:

1. The tests in this section must be run only if the DUT supports Class B
2. It is assumed the end-device has successfully performed beacon acquisition before the temporary Class B multicast session starts
 - TCL sends a BeaconFreqReq MAC command to the DUT with Frequency set to any allowed channel frequency
 - DUT responds with a successful BeaconFreqAns

- 493 ○ TCL creates a multicast session on the DUT using McGroupSetupReq with the
- 494 following parameters:
- 495 ▪ McGroupID = 0
- 496 ▪ McAddr = 0x00ABCDEF
- 497 ▪ McKey = 0x0102030405060708090A0B0C0D0E0F10
- 498 ▪ minMcFCount = 0
- 499 ▪ maxMcFCount = 0x0000FFFF
- 500 ○ DUT must respond with the McGroupSetupAns command without any errors, with
- 501 the correct McGroupID.
- 502 ○ TCL sends a McGroupStatusReq command.
- 503 ○ The DUT must respond correctly with the McGroupStatusAns command.
- 504 ○ TCL sends the McClassBSessionReq command to the DUT to setup a temporary Class
- 505 B multicast session with
- 506 ▪ McGroupID = 0
- 507 ▪ SessionTime = 3*128 seconds from the current time
- 508 ▪ Periodicity = 4
- 509 ▪ TimeOut = 256 seconds
- 510 ▪ DLFreq = Any allowed frequency (refer to Regional Parameters Specification)
- 511 ▪ DR = Max125KHzDR or Max500kHzDR (refer to Regional Parameters
- 512 Specification)
- 513 ○ The DUT must respond with McClassBSessionAns command with
- 514 ▪ McGroupID = 0
- 515 ▪ DRError = 0
- 516 ▪ FreqError = 0
- 517 ▪ McGroupUndefined = 0
- 518 ▪ StartMissed = 0
- 519 ▪ Reserved for future errors = 0
- 520 ▪ TimeToStart = the number of seconds from the McClassBSessionAns uplink to
- 521 the beginning of the multicast session
- 522 ○ TCL sends CP-CMD RxAppCntReq to DUT
- 523 ○ DUT must respond with RxAppCntAns
- 524 ○ After the SessionTime, TCL sends repeated CP-CMD TxFramesCtrlReq with
- 525 FrameType=0 (No change) using multicast ping-slots, multicast address and session
- 526 context.
- 527 ○ Wait until SessionTimeOut.
- 528 ○ TCL sends CP-CMD RxAppCntReq to DUT.
- 529 ○ DUT must respond with RxAppCntAns with the RxAppCnt field incremented by the
- 530 number of downlinks sent by the TCL. (Tolerance of 5%)
- 531
- 532 ○ The TCL sends 3 Unconfirmed frames using multicast ping slots, multicast address and
- 533 session context after the multicast session has closed.

- 534 ○ DUT must ignore the downlinks.
- 535 ○ TCL deletes the McGroupID = 0 and creates a new group with McGroupID = 0 again,
- 536 with minMcFCnt = 5 and maxMcFCnt = 20
- 537 ○ TCL creates a new McClassBSession
- 538 ○ After the Session time starts, TCL sends Unconfirmed frame using multicast and
- 539 session context with McFCnt < minMcFCnt.
- 540 ○ DUT must ignore this downlink.
- 541 ○ TCL sends Unconfirmed frame using multicast and session context with McFCnt >
- 542 maxMcFCnt.
- 543 ○ DUT must ignore this downlink.
- 544 ○ TCL sends Confirmed frame using multicast and session context with McFCnt >
- 545 minMcFCnt and McFCnt > maxMcFCnt.
- 546 ○ DUT must ignore this downlink.
- 547 ○ TCL sends Unconfirmed MAC-CMD frame using multicast ping slot, multicast address
- 548 and session context.
- 549 ○ DUT must ignore this downlink.
- 550 ○ TCL sends Unconfirmed McGroupSetupReq frame using multicast ping slot, multicast
- 551 address and session context.
- 552 ○ DUT must ignore this downlink.

553

554 **With multiple groups**

- 555 • TCL sends McGroupSetupReq to create 2 McGroups ID = 0 and 1.
- 556 • DUT responds with McGroupSetupAns successfully.
- 557 • TCL sends McClassBSessionReq to create 2 sessions for each McGroupID 0 and 1 with
- 558 different parameters.
- 559 • DUT responds with McClassBSessionAns successfully.
- 560 • TCL changes the Periodicity of uplinks to 240 seconds.
- 561 • Wait for SessionTime.
- 562 • TCL sends repeated downlinks alternatively on McGroupID 0 and 1.
- 563 • TCL reverts the Periodicity of uplinks to 5 seconds.
- 564 • TCL checks the counter using CP-CMD RxAppCntAns to confirm that the downlinks are
- 565 received on both the multicast groups with a tolerance of 5%.
- 566 • TCL deletes McGroupID = 1 using McGroupDeleteReq command.
- 567 • DUT responds with McGroupDeleteAns command.

568

569 **With incorrect parameters**

- 570 • TCL sends the McClassBSessionReq command to the DUT to setup a temporary Class
- 571 B multicast session with
 - 572 ○ McGroupID = 1 (invalid)
 - 573 ○ SessionTime = 3*128 seconds from the current time

- 574 ○ Periodicity = 0
- 575 ○ Timeout = 32 seconds
- 576 ○ DLFreq = Any allowed frequency (refer to Regional Parameters
- 577 Specification)
- 578 ○ DR = Max125KHzDR or Max500kHzDR (refer to Regional Parameters
- 579 Specification)
- 580 • The DUT must respond with McClassBSessionAns command with
- 581 ○ McGroupID = 1
- 582 ○ DRError = 0
- 583 ○ FreqError = 0
- 584 ○ McGroupUndefined = 1
- 585 ○ StartMissed = 0
- 586 ○ Reserved for future errors = 0
- 587 • TCL sends the McClassBSessionReq command to the DUT to setup a temporary Class
- 588 B multicast session with
- 589 ○ McGroupID = 0
- 590 ○ SessionTime = Before the current time (invalid)
- 591 ○ Periodicity = 0
- 592 ○ Timeout = 32 seconds
- 593 ○ DLFreq = Any allowed frequency (refer to Regional Parameters
- 594 Specification)
- 595 ○ DR = Max125KHzDR or Max500kHzDR (refer to Regional Parameters
- 596 Specification)
- 597 • The DUT must respond with McClassBSessionAns command with
- 598 ○ McGroupID = 0
- 599 ○ DRError = 0
- 600 ○ FreqError = 0
- 601 ○ McGroupUndefined = 0
- 602 ○ StartMissed = 1
- 603 ○ Reserved for future errors = 0
- 604 • TCL sends the McClassBSessionReq command to the DUT to setup a temporary Class
- 605 B multicast session with
- 606 ○ McGroupID = 0
- 607 ○ SessionTime = 3*128 seconds from the current time
- 608 ○ Periodicity = 0
- 609 ○ Timeout = 32 seconds
- 610 ○ DLFreq = Invalid frequency (invalid)
- 611 ○ DR = Max125KHzDR or Max500kHzDR (refer to Regional Parameters
- 612 Specification)
- 613 • The DUT must respond with McClassBSessionAns command with
- 614 ○ McGroupID = 0

- 615 ○ DRError = 0
- 616 ○ FreqError = 1
- 617 ○ McGroupUndefined = 0
- 618 ○ StartMissed = 0
- 619 ○ Reserved for future errors = 0
- 620 • TCL sends the McClassBSessionReq command to the DUT to setup a temporary Class
- 621 B multicast session with
- 622 ○ McGroupID = 0
- 623 ○ SessionTime = 3*128 seconds from the current time
- 624 ○ Periodicity = 0
- 625 ○ TimeOut = 32 seconds
- 626 ○ DLFreq = Any allowed frequency (refer to Regional Parameters
- 627 Specification) (invalid)
- 628 ○ DR = Invalid DR
- 629 • The DUT must respond with McClassBSessionAns command with
- 630 ○ McGroupID = 0
- 631 ○ DRError = 1
- 632 ○ FreqError = 0
- 633 ○ McGroupUndefined = 0
- 634 ○ StartMissed = 0
- 635 ○ Reserved for future errors = 0
- 636 • TCL sends the McClassBSessionReq command to the DUT to setup a temporary Class
- 637 B multicast session with
- 638 ○ McGroupID = 0
- 639 ○ SessionTime = Current time + 16777215 + 20 seconds
- 640 ○ Periodicity = 0
- 641 ○ TimeOut = 32 seconds
- 642 ○ DLFreq = Any allowed frequency (refer to Regional Parameters
- 643 Specification)
- 644 ○ DR = Max125KHzDR or Max500kHzDR (refer to Regional Parameters
- 645 Specification)
- 646 • The DUT must respond with McClassBSessionAns command with
- 647 ○ McGroupID = 0
- 648 ○ DRError = 0
- 649 ○ FreqError = 0
- 650 ○ McGroupUndefined = 0
- 651 ○ StartMissed = 0
- 652 ○ Reserved for future errors = 0
- 653 ○ TimeToStart = 0xFFFFF
- 654



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655 Test Procedure Frame Sequence Chart

656

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		For device clock sync
	TCL sends Unconfirmed frame	←	CP-CMD DeviceTimeReq FPort = 224 Payload = [0x]21	
2	DUT sends Unconfirmed frame	→	MAC-CMD DeviceTimeReq Payload = [0x]0D	
	The TCL sends Unconfirmed frame	←	MAC-CMD DeviceTimeAns Payload = [0x]0DXXXXXXXXXX	
3	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD PackageVersionReq Payload = [0x]00	
4	DUT sends Unconfirmed frame	→	CMD PackageVersionAns Package Identifier = 2 Package Version = 2 Payload = [0x]00XXXX	
	The TCL sends Unconfirmed frame	←	CMD McGroupSetupReq McGroupID = 0 McAddr = 0x00ABCDEF McKey_encrypted = aes128_decrypt (McKEKey, McKey), where McKey = 0x0102030405060708090A0B0C0D 0E0F10 minMcFCount = 0 maxMcFCount = 0x0000FFFF Payload = [0x]02 XX(for 29 octets)	
5	DUT sends Unconfirmed frame	→	CMD McGroupSetupAns IDerror = 0 McGroupID = 0 Payload = [0x]0200	
	The TCL sends Unconfirmed frame	←	CMD McGroupStatusReq ReqGroupMask = 1 Payload = [0x]0101	
6	DUT sends Unconfirmed frame	→	CMD McGroupStatusAns AnsGroupMask = 1 NbTotalGroups = 1 McGroupID = 0 + McAddr = 0x00ABCDEF Payload = [0x]0111McGroupID+McAddr	
	The TCL sends Unconfirmed frame	←	CMD McGroupDeleteReq McGroupID = 0 Payload = [0x]0300	
7	DUT sends Unconfirmed frame	→	CMD McGroupDeleteAns McGroupUndefined = 0 McGroupID = 0 Payload = [0x]0300	

	The TCL sends Unconfirmed frame	←	CMD McGroupStatusReq ReqGroupMask = 1 Payload = [0x]0101	
8	DUT sends Unconfirmed frame	→	CMD McGroupStatusAns AnsGroupMask = 0 NbTotalGroups = 0 Payload = [0x]0100	
	The TCL sends Unconfirmed frame	←	CMD McGroupSetupReq McGroupID = 0 McAddr = 0x00ABCDEF McKey_encrypted = aes128_decrypt (McKEKey, McKey), where McKey = 0x0102030405060708090A0B0C0D 0E0F10 minMcFCount = 0 maxMcFCount = 0x0000FFFF Payload = [0x]02 XX(for 29 octets)	
9	DUT sends Unconfirmed frame	→	CMD McGroupSetupAns IDerror = 0 McGroupID = 0 Payload = [0x]0200	
	The TCL sends Unconfirmed frame	←	CMD McGroupSetupReq McGroupID = 1 McAddr = 0x00123456 McKey_encrypted = aes128_decrypt (McKEKey, McKey), where McKey = 0x100F0E0D0C0B0A09080706050 4030201 minMcFCount = 0 maxMcFCount = 0x0000FFFF Payload = [0x]02 XX(for 29 octets)	
10	If the DUT does not support multiple multicast groups, skip to the section- status of non-existing group after this uplink If the DUT supports multicast groups, DUT sends Unconfirmed frame	→	If the DUT does not support multiple multicast groups CMD McGroupSetupAns IDerror = 1 McGroupID = 1 Payload = [0x]02XX If the DUT supports multiple multicast groups CMD McGroupSetupAns IDerror = 0 McGroupID = 1 Payload = [0x]02XX	
	The TCL sends Unconfirmed frame	←	CMD McGroupStatusReq ReqGroupMask = 0x3 Payload = [0x]0103	

11	DUT sends Unconfirmed frame	→	CMD McGroupStatusAns AnsGroupMask = 0x3 NbTotalGroups = 2 McGroupID = 0 + McAddr = 0x00ABCDEF [McGroupID = 1 + McAddr = 0x00123456] Payload = [0x]01XX McGroupID+McAddr for both groups	
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = lowest DR, refer [2] Payload = [0x]03XXXXXXXXXX	
12	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	
	The TCL sends Unconfirmed frame	←	CMD McGroupStatusReq ReqGroupMask = 0x3 Payload = [0x]0103	
13	DUT sends Unconfirmed frame	→	<u>For DC plan devices only</u> CMD McGroupStatusAns NbTotalGroups = 2 AnsGroupMask = 0x03 McGroupID + McAddr for both groups Payload = [0x]01XX[McGroupID + McAddr for both groups] <u>For FC plan devices only</u> CMD McGroupStatusAns NbTotalGroups = 2 AnsGroupMask = 0x01 McGroupID + McAddr for first group Payload = [0x]01XX[McGroupID + McAddr for first group]	
	The TCL sends Unconfirmed frame	←	CMD McGroupStatusReq ReqGroupMask = 0x2 Payload = [0x]0102	
14	DUT sends Unconfirmed frame	→	CMD McGroupStatusAns AnsGroupMask = 0x2 NbTotalGroups = 2 [McGroupID + McAddr] for second group Payload = [0x]01XX[McGroupID+McAddr for second group]	
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = max125kHzDR, refer [2] Payload = [0x]03XXXXXXXXXX	
15	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	

	The TCL sends Unconfirmed frame	←	CMD McGroupDeleteReq McGroupID = 1 Payload = [0x]03XX	
16	DUT sends Unconfirmed frame	→	CMD McGroupDeleteAns McGroupUndefined = 0 McGroupID = 1 Payload = [0x]03XX	
	The TCL sends Unconfirmed frame	←	CMD McGroupStatusReq ReqGroupMask = 0x3 Payload = [0x]0103	
17	DUT sends Unconfirmed frame	→	CMD McGroupStatusAns AnsGroupMask = 0x1 NbTotalGroups = 1 McGroupID = 0 + McAddr = 0x00ABCDEF Payload = [0x]01XX McGroupID+McAddr for the second group	
Status and Delete of non-existing group				
18	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD McGroupDeleteReq McGroupID = 0 Payload = [0x]03XX	
19	DUT sends Unconfirmed frame	→	CMD McGroupDeleteAns McGroupUndefined = 0 McGroupID = 0 Payload = [0x]03XX	
	The TCL sends Unconfirmed frame	←	CMD McGroupStatusReq ReqGroupMask = 0x8 Payload = [0x]01XX	
20	DUT sends Unconfirmed frame	→	CMD McGroupStatusAns AnsGroupMask = 0 NbTotalGroups = 0 Payload = [0x]0100	
	The TCL sends Unconfirmed frame	←	CMD McGroupDeleteReq McGroupID = 1 Payload = [0x]03XX	
21	DUT sends Unconfirmed frame	→	CMD McGroupDeleteAns McGroupUndefined = 1 McGroupID = 1 Payload = [0x]03XX	
McClassCSessionReq				
22	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD McGroupSetupReq McGroupID = 0 McAddr = 0x00ABCDEF McKey = 0x0102030405060708090A0B0C0D 0E0F10 minMcFCount = 0 maxMcFCount = 0x0000FFFF Payload = [0x]02 XX(for 29 octets)	

23	DUT sends Unconfirmed frame	→	CMD McGroupSetupAns IDerror = 0 McGroupID = 0 Payload = [0x]02XX	
	TCL sends Unconfirmed frame on the RX2 window	←	CMD McGroupStatusReq ReqGroupMask = 1 Payload = [0x]01XX	
24	DUT sends Unconfirmed frame	→	CMD McGroupStatusAns AnsGroupMask = 1 NbTotalGroups = 1 Payload = [0x]01XX	
	The TCL sends Unconfirmed frame	←	CMD McClassCSessionReq McGroupID = 0 SessionTime = 30 seconds from the current time SessionTimeOut = 32 seconds DLFreq = Any allowed frequency other than the default RX2 frequency, refer [2] DR = Max125KHzDR or Max500kHzDR, refer [2] Payload = [0x]04 XX(for 10 octets)	
25	DUT sends Unconfirmed frame	→	CMD McClassCSessionAns McGroupID = 0 DRError = 0 FreqError = 0 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 TimeToStart = number of seconds from the McClassCSessionAns uplink to the beginning of the multicast session Payload = [0x]04XXXXXXXX	
	The TCL sends Unconfirmed frames	←	CP-CMD RxAppCntReq FPort = 224 Payload = [0x]09	
26	DUT sends Unconfirmed frame	→	CP-CMD RxAppCntAns FPort = 224 Payload = [0x]09XXXX RxAppCnt = x	
	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 120s Payload = [0x]0608	Uplink Periodicity set to 120s
	Wait for SessionTime			

27	The TCL sends Unconfirmed frame using multicast and session context during the RXC (3) window 500ms after the gateway completes the transmission of the previous downlink. i.e. Downlink TimeonAir + 500ms Repeat 20 times	← R[20]	CP-CMD TxFramesCtrlReq FPort 224 Frame Type = No change Payload [0x]0700	
	Wait until SessionTimeout			
28	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 5s Payload = [0x]0601	Uplink Periodicity set to 5s
29	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frames	←	CP-CMD RxAppCntReq FPort = 224 Payload = [0x]09	
30	DUT sends Unconfirmed frame	→	CP-CMD RxAppCntAns FPort = 224 Payload = [0x]09XXXX RxAppCnt ≥ x + 19 + 3 = y (tolerance of 5% included)	
31	The TCL sends Unconfirmed frame using multicast and session context during the RXC (3) window 3s after the uplink is received and for the repeat downlinks, 500ms after the gateway completes the transmission of the previous downlink. i.e. Downlink TimeonAir + 500ms	← R [3]	CP-CMD TxFramesCtrlReq FPort 224 Frame Type = No change Payload [0x]0700	Multicast session is not open after Timeout
32	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frames	←	CP-CMD RxAppCntReq FPort = 224 Payload = [0x]09	
33	DUT sends Unconfirmed frame	→	CP-CMD RxAppCntAns FPort = 224 Payload = [0x]09XXXX RxAppCnt = y + 1	
	The TCL sends Unconfirmed frame	←	CMD McGroupDeleteReq McGroupID = 0 Payload = [0x]03XX	
34	DUT sends Unconfirmed frame	→	CMD McGroupDeleteAns McGroupUndefined = 0 McGroupID = 0 Payload = [0x]03XX	

	The TCL sends Unconfirmed frame	←	CMD McGroupSetupReq McGroupID = 0 McAddr = 0x00ABCDEF McKey = 0x0102030405060708090A0B0C0D 0E0F10 minMcFCnt = 5 maxMcFCnt = 20 Payload = [0x]02 XX(for 29 octets)	
35	DUT sends Unconfirmed frame	→	CMD McGroupSetupAns IDerror = 0 McGroupID = 0 Payload = [0x]02XX	
	The TCL sends Unconfirmed frame	←	CMD McClassCSessionReq McGroupID = 0 SessionTime = 30 seconds from the current time SessionTimeOut = 32 seconds DLFreq = Any allowed frequency other than the default RX2 frequency, refer [2] DR = Max125KHzDR or Max500kHzDR, refer [2] Payload = [0x]04 XX(for 10 octets)	
36	DUT sends Unconfirmed frame	→	CMD McClassCSessionAns McGroupID = 0 DRError = 0 FreqError = 0 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 TimeToStart = number of seconds from the McClassCSessionAns uplink to the beginning of the multicast session Payload = [0x]04XXXXXXXX	
	The TCL sends Unconfirmed frames	←	CP-CMD RxAppCntReq FPort = 224 Payload = [0x]09	
37	DUT sends Unconfirmed frame	→	CP-CMD RxAppCntAns FPort = 224 Payload = [0x]09XXXX RxAppCnt = z	
	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 50s Payload = [0x]0606	Uplink Periodicity set to 50s
	Wait for SessionTime			
38	TCL sends Unconfirmed frame using multicast and session context during the RXC (3) window McFCnt = 4	← R[2]	CP-CMD TxFramesCtrlReq FPort 224 Frame Type = No change Payload [0x]0700	McFCnt < minMcFCnt

39	TCL sends Unconfirmed frame using multicast and session context during the RXC (3) window 500ms after the gateway completes the transmission of the previous downlink. i.e. Downlink TimeonAir + 500ms McFCnt = 21	← R[2]	CP-CMD TxFramesCtrlReq FPort 224 Frame Type = No change Payload [0x]0700	McFCnt > maxMcFCnt
40	The TCL sends Confirmed frame using multicast and session context during the RXC (3) window 500ms after the gateway completes the transmission of the previous downlink. i.e. Downlink TimeonAir + 500ms McFCnt = 6	← R[2]	CP-CMD TxFramesCtrlReq FPort 224 Frame Type = No change Payload [0x]0700	
41	DUT sends Unconfirmed frame	→	No ACK is sent	
42	The TCL sends Unconfirmed frame using multicast and session context during the RXC (3) window 3s after the uplink is received McFCnt = 7	←	MAC-CMD LinkADRReq DataRate = lowest DR, refer [2] Payload = [0x]03XXXXXXXX	
43	DUT sends Unconfirmed frame	→	No LinkADRAAns must be sent	
44	The TCL sends Unconfirmed frame using multicast and session context during the RXC (3) window 3s after the uplink is received McFCnt = 8	←	CMD McGroupSetupReq McGroupID = 0 McAddr = 0x00ABCDEF McKey = 0x0102030405060708090A0B0C0D 0E0F10 minMcFCount = 0 maxMcFCount = 0x0000FFFF Payload = [0x]02 XX(for 29 octets)	
45	DUT sends Unconfirmed frame	→	No McGroupSetupAns must be sent	
	Wait until SessionTimeout			
46	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 5s Payload = [0x]0601	Uplink Periodicity set to 5s
46-1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frames	←	CP-CMD RxAppCntReq FPort = 224 Payload = [0x]09	
46-2	DUT sends Unconfirmed frame	→	CP-CMD RxAppCntAns FPort = 224 Payload = [0x]09XXXX RxAppCnt = z + 2	
Multiple groups test (These tests must be executed only if the device supports Multiple groups for Class C multicast)				
47	DUT sends Unconfirmed frame	→		

	The TCL sends Unconfirmed frame	←	CMD McGroupSetupReq McGroupID = 0 McAddr = 0x00ABCDEF McKey_encrypted = aes128_decrypt (McKEKey, McKey), where McKey = 0x0102030405060708090A0B0C0D 0E0F10 minMcFCount = 0 maxMcFCount = 0x0000FFFF Payload = [0x]02 XX(for 29 octets)	
48	DUT sends Unconfirmed frame	→	CMD McGroupSetupAns IDerror = 0 McGroupID = 0 Payload = [0x]0200	
	The TCL sends Unconfirmed frame	←	CMD McGroupSetupReq McGroupID = 1 McAddr = 0x00123456 McKey_encrypted = aes128_decrypt (McKEKey, McKey), where McKey = 0x100F0E0D0C0B0A09080706050 4030201 minMcFCount = 0 maxMcFCount = 0x0000FFFF Payload = [0x]02 XX(for 29 octets)	
49	DUT sends Unconfirmed frame	→	CMD McGroupSetupAns IDerror = 0 McGroupID = 1 Payload = [0x]02XX	
	The TCL sends Unconfirmed frame	←	CMD McClassCSessionReq McGroupID = 0 SessionTime = 30 seconds from the current time SessionTimeOut = 128 seconds DLFreq = Any allowed frequency other than the default RX2 frequency, refer [2] DR = Max125KHzDR or Max500kHzDR, refer [2] Payload = [0x]04 XX(for 10 octets)	
50	DUT sends Unconfirmed frame	→	CMD McClassCSessionAns McGroupID = 0 DRError = 0 FreqError = 0 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 TimeToStart = number of seconds from the McClassCSessionAns uplink to the beginning of the multicast session Payload = [0x]04XXXXXXXX	

	The TCL sends Unconfirmed frame	←	CMD McClassCSessionReq McGroupID = 1 SessionTime = 25 seconds from the current time SessionTimeOut = 128 seconds DLFreq = Same as McGroupID 0 DR = Same as McGroupID 0 Payload = [0x]04 XX(for 10 octets)	
51	DUT sends Unconfirmed frame	→	CMD McClassCSessionAns McGroupID = 1 DRError = 0 FreqError = 0 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 TimeToStart = number of seconds from the McClassCSessionAns uplink to the beginning of the multicast session Payload = [0x]04XXXXXXXXXX	
	The TCL sends Unconfirmed frames	←	CP-CMD RxAppCntReq FPort = 224 Payload = [0x]09	
52	DUT sends Unconfirmed frame	→	CP-CMD RxAppCntAns FPort = 224 Payload = [0x]09XXXX RxAppCnt = x	
	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 240s Payload = [0x]0609	Uplink Periodicity set to 240s
	Wait for SessionTime			
53	The TCL sends Unconfirmed frame using multicast and session context for McGroupID = 0 during the RXC (3) window 500ms after the gateway completes the transmission of the previous downlink. i.e. Downlink TimeonAir + 500ms	← R[20]	CP-CMD TxFramesCtrlReq FPort 224 Frame Type = No change Payload [0x]0700	
	The TCL sends Unconfirmed frame using multicast and session context for McGroupID = 1 during the RXC (3) window 500ms after the gateway completes the transmission of the previous downlink. i.e. Downlink TimeonAir + 500ms		CP-CMD TxFramesCtrlReq FPort 224 Frame Type = No change Payload [0x]0700	
	Alternatively repeat the downlinks 20 times for both McGroupID = 0 and McGroup = 1			
54	Wait until SessionTimeOut			
55	DUT sends Unconfirmed frame	→		

	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 5s Payload = [0x]0601	Uplink Periodicity set to 5s
56	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frames	←	CP-CMD RxAppCntReq FPort = 224 Payload = [0x]09	
57	DUT sends Unconfirmed frame	→	CP-CMD RxAppCntAns FPort = 224 Payload = [0x]09XXXX RxAppCnt $\geq x + 38 + 3 = y$ (tolerance of 5% included)	
<i>Multiple group test ends here</i>				
	The TCL sends Unconfirmed frame	←	CMD McGroupDeleteReq McGroupID = 1 Payload = [0x]0300	
58	DUT sends Unconfirmed frame	→	CMD McGroupDeleteAns McGroupUndefined = 0 McGroupID = 1 Payload = [0x]03XX	
	The TCL sends Unconfirmed frame	←	CMD McClassCSessionReq McGroupID = 1 (invalid) SessionTime = 8 seconds from the current time SessionTimeOut = 32 seconds DLFreq = Any allowed frequency other than the default RX2 frequency, refer [2] DR = Max125KHzDR or Max500kHzDR, refer [2] Payload = [0x]04 XX(for 10 octets)	With incorrect parameters
59	DUT sends Unconfirmed frame	→	CMD McClassCSessionAns McGroupID = 1 DRError = 0 FreqError = 0 McGroupUndefined = 1 StartMissed = 0 Reserved for future errors = 0 Payload = [0x]04XX	
	The TCL sends Unconfirmed frame	←	CMD McClassCSessionReq McGroupID = 0 SessionTime = Before the current time (invalid) SessionTimeOut = 32 seconds DLFreq = Any allowed frequency other than the default RX2 frequency, refer [2] DR = Max125KHzDR or Max500kHzDR, refer [2] Payload = [0x]04 XX(for 10 octets)	

60	DUT sends Unconfirmed frame	→	CMD McClassCSessionAns McGroupID = 0 DRError = 0 FreqError = 0 McGroupUndefined = 0 StartMissed = 1 Reserved for future errors = 0 Payload = [0x]04XX	
	The TCL sends Unconfirmed frame	←	CMD McClassCSessionReq McGroupID = 0 SessionTime = 8 seconds from the current time SessionTimeout = 32 seconds DLFreq = Any frequency NOT allowed, refer [2] (invalid) DR = Max125KHzDR or Max500kHzDR, refer [2] Payload = [0x]04 XX(for 10 octets)	
61	DUT sends Unconfirmed frame	→	CMD McClassCSessionAns McGroupID = 0 DRError = 0 FreqError = 1 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 Payload = [0x]04XX	
	The TCL sends Unconfirmed frame	←	CMD McClassCSessionReq McGroupID = 0 SessionTime = 8 seconds from the current time SessionTimeout = 32 seconds DLFreq = Any allowed frequency other than the default RX2 frequency, refer [2] DR = Any DR NOT allowed, refer [2] Payload = [0x]04 XX(for 10 octets)	
62	DUT sends Unconfirmed frame	→	CMD McClassCSessionAns McGroupID = 0 DRError = 1 FreqError = 0 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 Payload = [0x]04XX	

	The TCL sends Unconfirmed frame	←	CMD McClassCSessionReq McGroupID = 0) SessionTime = Current time + 16777215 + 20 seconds SessionTimeout = 32 seconds DLFreq = Any allowed frequency other than the default RX2 frequency, refer [2] DR = Max125KHzDR or Max500kHzDR, refer [2] Payload = [0x]04 XX(for 10 octets)	Additional 20 seconds added to ensure that the 16777215 limit is fulfilled.
63	DUT sends Unconfirmed frame	→	CMD McClassCSessionAns McGroupID = 0 DRError = 0 FreqError = 0 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 TimeToStart = 0xFFFFFFFF Payload = [0x]04XXXXXXXXXX	
	The TCL sends Unconfirmed frame	←	CMD McGroupDeleteReq McGroupID = 0 Payload = [0x]03XX	
64	DUT sends Unconfirmed frame	→	CMD McGroupDeleteAns McGroupUndefined = 0 McGroupID = 0 Payload = [0x]03XX	
McClassBSessionReq				
Note: It is assumed the end-device has successfully performed beacon acquisition before the temporary Class B multicast session starts				
	TCL continuously sends beacons	← R	BCNPPayload	
65	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD McGroupSetupReq McGroupID = 0 McAddr = 0x00ABCDEF McKey_encrypted = aes128_decrypt (McKEKey, McKey), where McKey = 0x0102030405060708090A0B0C0D0E0F10 minMcFCount = 0 maxMcFCount = 0x0000FFFF Payload = [0x]02 XX(for 29 octets)	
66	DUT sends Unconfirmed frame	→	CMD McGroupSetupAns IDerror = 0 McGroupID = 0 Payload = [0x]02XX	
	The TCL sends Unconfirmed frame	←	CMD McGroupStatusReq ReqGroupMask = 1 Payload = [0x]01XX	

67	DUT sends Unconfirmed frame	→	CMD McGroupStatusAns AnsGroupMask = 1 NbTotalGroups = 1 Payload = [0x]01XX	
	The TCL sends Unconfirmed frame	←	CMD McClassBSessionReq McGroupID = 0 SessionTime = 3*128 seconds from current time Periodicity = 2 TimeOut = 128 seconds DLFreq = Any allowed frequency, refer [2] DR = Max125KHzDR or Max500kHzDR, refer [2] Payload = [0x]05 XX(for 10 octets)	
68	DUT sends Unconfirmed frame	→	CMD McClassBSessionAns McGroupID = 0 DRError = 0 FreqError = 0 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 TimeToStart = the number of seconds from the McClassBSessionAns uplink to the beginning of the multicast session Payload = [0x]05XXXXXXXXXX	
	The TCL sends Unconfirmed frames	←	CP-CMD RxAppCntReq FPort = 224 Payload = [0x]09	
69	DUT sends Unconfirmed frame	→	CP-CMD RxAppCntAns FPort = 224 Payload = [0x]09XXXX RxAppCnt = x	
	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 120s Payload = [0x]0608	Uplink Periodicity set to 120s
	Wait for SessionTime			
70	Wait for the DUT to send an Unconfirmed frame.	→		
71	The TCL sends Unconfirmed frame using multicast ping-slots, multicast address and session context Repeat 20 times	← R[20]	CP-CMD TxFramesCtrlReq FPort 224 Frame Type = No change Payload [0x]0700	
	Wait until SessionTimeOut			
72	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 5s Payload = [0x]0601	Uplink Periodicity set to 5s

73	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frames	←	CP-CMD RxAppCntReq FPort = 224 Payload = [0x]09	
74	DUT sends Unconfirmed frame	→	CP-CMD RxAppCntAns FPort = 224 Payload = [0x]09XXXX RxAppCnt = $x + 19 + 2 = y$ (tolerance of 5% included)	
75	The TCL sends Unconfirmed frame using multicast ping-slots, multicast address and session context	← R [3]	CP-CMD TxFramesCtrlReq FPort 224 Frame Type = No change Payload [0x]0700	Multicast session is not open after TimeOut
76	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frames	←	CP-CMD RxAppCntReq FPort = 224 Payload = [0x]09	
77	DUT sends Unconfirmed frame	→	CP-CMD RxAppCntAns FPort = 224 Payload = [0x]09XXXX RxAppCnt = $y + 1$	
	The TCL sends Unconfirmed frame	←	CMD McGroupDeleteReq McGroupID = 0 Payload = [0x]03XX	
78	DUT sends Unconfirmed frame	→	CMD McGroupDeleteAns McGroupUndefined = 0 McGroupID = 0 Payload = [0x]03XX	
	The TCL sends Unconfirmed frame	←	CMD McGroupSetupReq McGroupID = 0 McAddr = 0x00ABCDEF McKey = 0x0102030405060708090A0B0C0D0E0F10 minMcFCnt = 5 maxMcFCnt = 20 Payload = [0x]02 XX(for 29 octets)	
79	DUT sends Unconfirmed frame	→	CMD McGroupSetupAns IDerror = 0 McGroupID = 0 Payload = [0x]02XX	
	The TCL sends Unconfirmed frame	←	CMD McClassBSessionReq McGroupID = 0 SessionTime = 3*128 seconds from current time Periodicity = 2 TimeOut = 256 seconds DLFreq = Any allowed frequency, refer [2] DR = Max125KHzDR or Max500kHzDR, refer [2] Payload = [0x]05 XX(for 10 octets)	

80	DUT sends Unconfirmed frame	→	CMD McClassBSessionAns McGroupID = 0 DRError = 0 FreqError = 0 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 TimeToStart = the number of seconds from the McClassBSessionAns uplink to the beginning of the multicast session Payload = [0x]05XXXXXXXX	
	The TCL sends Unconfirmed frames	←	CP-CMD RxAppCntReq FPort = 224 Payload = [0x]09	
81	DUT sends Unconfirmed frame	→	CP-CMD RxAppCntAns FPort = 224 Payload = [0x]09XXXX RxAppCnt = z	
	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 120s Payload = [0x]0608	Uplink Periodicity set to 120s
82	Wait for SessionTime			
83	TCL sends Unconfirmed frame using multicast ping-slots, multicast address and session context	← R[2]	CP-CMD TxFramesCtrlReq FPort 224 Frame Type = No change Payload [0x]0700	McFCnt < minMcFCnt
	McFCnt = 4			
84	TCL sends Unconfirmed frame using multicast ping-slots, multicast address and session context	← R[2]	CP-CMD TxFramesCtrlReq FPort 224 Frame Type = No change Payload [0x]0700	McFCnt > maxMcFCnt
	McFCnt = 21			
85	The TCL sends Confirmed frame using multicast ping-slots, multicast address and session context	← R[2]	CP-CMD TxFramesCtrlReq FPort 224 Frame Type = No change Payload [0x]0700	
86	DUT sends Unconfirmed frame	→	No ACK is sent	
87	The TCL sends Unconfirmed frame using multicast ping-slots, multicast address and session context	←	MAC-CMD LinkADRRReq DataRate = lowest DR, refer [2] Payload = [0x]03XXXXXXXX	
	McFCnt = 7			
88	DUT sends Unconfirmed frame	→	No LinkADRAns must be sent	

89	The TCL sends Unconfirmed frame using multicast ping-slots, multicast address and session context McFCnt = 8	←	CMD McGroupSetupReq McGroupID = 0 McAddr = 0x00ABCDEF McKey = 0x0102030405060708090A0B0C0D 0E0F10 minMcFCount = 0 maxMcFCount = 0x0000FFFF Payload = [0x]02 XX(for 29 octets)	
90	DUT sends Unconfirmed frame	→	No McGroupSetupAns must be sent	
	Wait until SessionTimeout			
91	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 5s Payload = [0x]0601	Uplink Periodicity set to 5s
92	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frames	←	CP-CMD RxAppCntReq FPort = 224 Payload = [0x]09	
93	DUT sends Unconfirmed frame	→	CP-CMD RxAppCntAns FPort = 224 Payload = [0x]09XXXX RxAppCnt = z + 4	
Multiple groups test (These tests must be executed only if the device supports Multiple groups for Class B multicast)				
94	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD McGroupSetupReq McGroupID = 0 McAddr = 0x00ABCDEF McKey_encrypted = aes128_decrypt (McKEKey, McKey), where McKey = 0x0102030405060708090A0B0C0D 0E0F10 minMcFCount = 0 maxMcFCount = 0x0000FFFF Payload = [0x]02 XX(for 29 octets)	
95	DUT sends Unconfirmed frame	→	CMD McGroupSetupAns IDerror = 0 McGroupID = 0 Payload = [0x]02XX	

	The TCL sends Unconfirmed frame	←	CMD McGroupSetupReq McGroupID = 1 McAddr = 0x00123456 McKey_encrypted = aes128_decrypt (McKEKey, McKey), where McKey = 0x100F0E0D0C0B0A09080706050 4030201 minMcFCount = 0 maxMcFCount = 0x0000FFFF Payload = [0x]02 XX(for 29 octets)	
96	DUT sends Unconfirmed frame	→	CMD McGroupSetupAns IDerror = 0 McGroupID = 1 Payload = [0x]02XX	
	The TCL sends Unconfirmed frame	←	CMD McClassBSessionReq McGroupID = 0 SessionTime = 3*128 seconds from current time Periodicity = 1 sec TimeOut = 128 seconds DLFreq = Any allowed frequency, refer [2] DR = Max125KHzDR or Max500kHzDR, refer [2] Payload = [0x]05 XX(for 10 octets)	
97	DUT sends Unconfirmed frame	→	CMD McClassBSessionAns McGroupID = 0 DRError = 0 FreqError = 0 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 TimeToStart = the number of seconds from the McClassBSessionAns uplink to the beginning of the multicast session Payload = [0x]05XXXXXXXX	
	The TCL sends Unconfirmed frame	←	CMD McClassBSessionReq McGroupID = 1 SessionTime = 3*128 seconds from current time Periodicity = 2 sec TimeOut = 128 seconds DLFreq = Any frequency other than default for Rx2, refer [2] DR = Max125KHzDR-1 or Max500kHzDR-1, refer [2] Payload = [0x]05 XX(for 10 octets)	

98	DUT sends Unconfirmed frame	→	CMD McClassBSessionAns McGroupID = 1 DRError = 0 FreqError = 0 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 TimeToStart = the number of seconds from the McClassBSessionAns uplink to the beginning of the multicast session Payload = [0x]05XXXXXXXX	
	The TCL sends Unconfirmed frames	←	CP-CMD RxAppCntReq FPort = 224 Payload = [0x]09	
99	DUT sends Unconfirmed frame	→	CP-CMD RxAppCntAns FPort = 224 Payload = [0x]09XXXX RxAppCnt = x	
	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 240s Payload = [0x]0609	Uplink Periodicity set to 240s
	Wait for SessionTime			
100	Wait for the DUT to send an Unconfirmed frame.	→		
101	The TCL sends Unconfirmed frame using multicast ping-slots, multicast address and session context for McGroupID = 0	← R[20]	CP-CMD TxFramesCtrlReq FPort 224 Frame Type = No change Payload [0x]0700	
	The TCL sends Unconfirmed frame using multicast ping-slots, multicast address and session context for McGroupID = 1		CP-CMD TxFramesCtrlReq FPort 224 Frame Type = No change Payload [0x]0700	
	Alternatively repeat the downlinks 20 times for both McGroupID = 0 and McGroup = 1			
102	Wait until SessionTimeOut			
103	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 5s Payload = [0x]0601	Uplink Periodicity set to 5s
104	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frames	←	CP-CMD RxAppCntReq FPort = 224 Payload = [0x]09	
105	DUT sends Unconfirmed frame	→	CP-CMD RxAppCntAns FPort = 224 Payload = [0x]09XXXX RxAppCnt = x + 38 + 2 (tolerance of 5% included)	
Multiple groups Class B test ends here				

	The TCL sends Unconfirmed frame	←	CMD McGroupDeleteReq McGroupID = 1 Payload = [0x]0301	
106	DUT sends Unconfirmed frame	→	CMD McGroupDeleteAns McGroupUndefined = 0 McGroupID = 1 Payload = [0x]03XX	
	The TCL sends Unconfirmed frame	←	CMD McClassBSessionReq McGroupID = 1 (invalid) SessionTime = 3*128 seconds from current time Periodicity = 0 TimeOut = 32 seconds DLFreq = Any allowed frequency, refer [2] DR = Max125KHzDR or Max500kHzDR, refer [2] Payload = [0x]05 XX(for 10 octets)	With incorrect parameters
107	DUT sends Unconfirmed frame	→	CMD McClassBSessionAns McGroupID = 1 DRError = 0 FreqError = 0 McGroupUndefined = 1 StartMissed = 0 Reserved for future errors = 0 Payload = [0x]05XX	
	The TCL sends Unconfirmed frame	←	CMD McClassBSessionReq McGroupID = 0 SessionTime = Before the current time Periodicity = 0 TimeOut = 32 seconds DLFreq = Any allowed frequency, refer [2] DR = Max125KHzDR or Max500kHzDR, refer [2] Payload = [0x]05 XX(for 10 octets)	
108	DUT sends Unconfirmed frame	→	CMD McClassBSessionAns McGroupID = 0 DRError = 0 FreqError = 0 McGroupUndefined = 0 StartMissed = 1 Reserved for future errors = 0 Payload = [0x]05XX	

	The TCL sends Unconfirmed frame	←	CMD McClassBSessionReq McGroupID = 0 SessionTime = 3*128 seconds from current time Periodicity = 0 TimeOut = 32 seconds DLFreq = Any frequency NOT allowed, refer [2] DR = Max125KHzDR or Max500kHzDR, refer [2] Payload = [0x]05 XX(for 10 octets)	
109	DUT sends Unconfirmed frame	→	CMD McClassBSessionAns McGroupID = 0 DRError = 0 FreqError = 1 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 Payload = [0x]05XX	
	The TCL sends Unconfirmed frame	←	CMD McClassBSessionReq McGroupID = 0 SessionTime = 3*128 seconds from current time Periodicity = 0 TimeOut = 32 seconds DLFreq = Any allowed frequency, refer [2] DR = Any DR NOT allowed, refer [2] Payload = [0x]05 XX(for 10 octets)	
110	DUT sends Unconfirmed frame	→	CMD McClassBSessionAns McGroupID = 0 DRError = 1 FreqError = 0 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 Payload = [0x]05XX	
	The TCL sends Unconfirmed frame	←	CMD McClassBSessionReq McGroupID = 0 SessionTime = Current time + 16777215 + 20 seconds Periodicity = 0 TimeOut = 32 seconds DLFreq = Any allowed frequency, refer [2] DR = Max125KHzDR or Max500kHzDR, refer [2] Payload = [0x]05 XX(for 10 octets)	Additional 20 seconds added to ensure that the 16777215 limit is fulfilled.

111	DUT sends Unconfirmed frame	→	CMD McClassBSessionAns McGroupID = 0 DRError = 0 FreqError = 0 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 TimeToStart = [0x]FFFFFF Payload = [0x]05XXXXXXX	
	The TCL sends Unconfirmed frame	←	CMD McGroupDeleteReq McGroupID = 0 Payload = [0x]03XX	
112	DUT sends Unconfirmed frame	→	CMD McGroupDeleteAns McGroupUndefined = 0 McGroupID = 0 Payload = [0x]03XX	

657

658 5. Fragmented Data Block Transport

659 Test Purpose: This test verifies all the Fragmented Data Block Transport commands from TS004-
660 Fragmented_Data_Block_Transport-v2.0.0 [9].

661

662 **Note:** It is recommended that a default FPort value of 201 be used to promote interoperability. The
663 default FPort value is obtained from the Certification Questionnaire.

664 PackageVersion

- 665 • TCL sends the command PackageVersionReq to the DUT
- 666 • DUT must respond with the command PackageVersionAns with Package Identifier = 3 and
667 Package Version = 2

668 Fragmentation

- 669 • TCL sends the CP-CMD SessionCntReq to retrieve the FragSessionCntPrev from the DUT
- 670 • DUT sends the SessionCntPrev
- 671 • TCL sends the command FragSessionSetupReq to the DUT with
 - 672 ○ FragSession
 - 673 ■ FragIndex = 0
 - 674 ■ McGroupBitMask = 0000
 - 675 ○ NbFrag = 6
 - 676 ○ FragSize = 170
 - 677 ○ Control
 - 678 ■ FragAlgo = 0
 - 679 ■ BlockAckDelay = 0
 - 680 ■ AckReception = 1
 - 681 ○ Padding = 25

- 682 ○ SessionCnt = SessionCnt[FragIndex]
- 683 ○ Descriptor = obtained from the Certification Questionnaire
- 684 ○ MIC = 0XXXXXXXX (depends on SessionCnt)
- 685 SessionCnt[FragIndex] += 1
- 686 • DUT must respond with the command FragSessionSetupAns with all bits set to 0
- 687 • TCL sends the 2nd, 3rd, 4th, 5th and 6th fragmented data block from a unicast address.
- 688 • TCL sends the FragSessionStatusReq to the DUT
- 689 • DUT must respond with the FragSessionStatusAns with NbFragReceived = total number of
- 690 fragments received and MissingFrag = number of fragments still required to reconstruct the
- 691 data block
- 692 • TCL sends a maximum of 4 redundancy fragments until the DUT sends the
- 693 FragDataBlockReceivedReq
- 694 • Once the data block is fully received, FragDataBlockReceivedReq must be sent with MICError
- 695 = 0
- 696 • TCL responds with FragDataBlockReceivedAns
- 697 • TCL sends the FragSessionStatusReq to the DUT
- 698 • DUT must not respond with the FragSessionStatusAns as the full data block has already been
- 699 received
- 700 • TCL sends FragSessionSetupReq with FragIndex = 0, SessionCnt = SessionCnt [FragIndex] and
- 701 an invalid MIC. SessionCnt [FragIndex] += 1.
- 702 • DUT must respond with FragSessionSetupAns
- 703 • TCL sends the 6 fragments and 4 redundancy frames until the DUT sends the
- 704 FragDataBlockReceivedReq
- 705 • TCL sends the FragDataBlockReceivedReq to the DUT with MICError = 1
- 706 • DUT must respond with the FragDataBlockReceivedAns
- 707 • TCL sends the command FragSessionSetupReq to the DUT for an incorrect SessionCnt.
- 708 • DUT must reply with FragSessionSetupAns with Session replay bit set to 1.
- 709 • TCL sends the FragSessionDeleteReq command to the DUT with FragIndex = 0
- 710 • DUT must respond with FragSessionDeleteAns command with the bit 2 set to 0 and the
- 711 FragIndex = 0
- 712 • TCL sends the FragSessionDeleteReq command to the DUT with FragIndex = 0 (already deleted
- 713 FragIndex)
- 714 • DUT must respond with FragSessionDeleteAns command with the bit 2 set to 1
- 715 • TCL sends the command FragSessionStatusReq to the DUT with FragIndex = 0 (already
- 716 deleted FragIndex)
- 717 • DUT must respond with FragSessionStatusAns with bit 2 set to 1 and must not include
- 718 ReceivedIndex and MissingFrag field.

719

720 Class B - Multicast and Fragmentation Setup

721 Note: If the device does not support multicast or Class B, skip to the next test.

- 722
- 723
- TCL continuously sends beacons
- 724
- TCL creates a multicast session on the DUT using McGroupSetupReq with the following
- 725
- parameters:
- McGroupID = 0
 - McAddr = 0x00ABCDEF
 - McKey = 0x0102030405060708090A0B0C0D0E0F10
 - minMcFCount = 0
 - maxMcFCount = 0x0000FFFF
- 726
- 727
- 728
- 729
- 730
- DUT must respond with the McGroupSetupAns command without any errors, with the correct
- 731
- McGroupID.
- TCL sends the McClassBSessionReq command to the DUT to setup a temporary Class B
- 732
- multicast session with
- McGroupID = 0
 - SessionTime = 3*128 seconds from the current time
 - Periodicity = 0
 - TimeOut = 128 seconds
 - DLFreq = Any allowed frequency (refer to Regional Parameters Specification)
 - DR = Max125KHzDR (refer to Regional Parameters Specification)
- 733
- 734
- The DUT must respond with McClassBSessionAns command with
- 735
- McGroupID = 0
 - DRError = 0
 - FreqError = 0
 - McGroupUndefined = 0
 - StartMissed = 0
 - Reserved for future errors = 0
 - TimeToStart = the number of seconds from the McClassBSessionAns uplink to the
- 736
- beginning of the multicast session
- TCL sends the command FragSessionSetupReq to the DUT with
- 737
- FragSession
- 738
- FragIndex = 0
- 739
- McGroupBitMask = 0000
- 740
- NbFrag = 6
- 741
- FragSize = 170
- 742
- Control
- 743
- FragAlgo = 0
- 744
- BlockAckDelay = 0
- 745
- AckReception = 1
- 746
- Padding = 25
- 747
- 748
- 749
- 750
- 751
- 752
- 753
- 754
- 755
- 756
- 757
- 758
- 759
- 760

- 761 • SessionCnt = SessionCnt[FragIndex]DUT must respond with the command
762 FragSessionSetupAns with all bits set to 0. The response must be sent after a delay of
763 $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink.
- 764 • TCL sets the Periodicity to 30 seconds
- 765 • After the SessionTime, TCL sends fragmented broadcast using multicast ping-slots, multicast
766 address and session context. TCL multicasts the 6 fragments and the 4 redundancy frames.
- 767 • DUT must ignore these multicast fragments as only unicast is allowed.
- 768 • TCL sends the command FragSessionStatusReq to the DUT with FragIndex = 0 and the
769 Participants = 1
- 770 • FragDataBlockReceivedReq is not sent by the DUT
- 771 • DUT must respond with FragSessionStatusAns with NbFragReceived = 0. The response must
772 be sent after a delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink.
- 773 • TCL sends the command FragSessionSetupReq to the DUT with
- 774 ○ FragSession
- 775 ▪ FragIndex = 0
- 776 ▪ McGroupBitMask = 0001
- 777 ○ NbFrag = 6
- 778 ○ FragSize = 170
- 779 ○ Control
- 780 ▪ FragAlgo = 0
- 781 ▪ BlockAckDelay = 0
- 782 ▪ AckReception = 1
- 783 ○ Padding = 25
- 784 ○ SessionCnt = SessionCnt[FragIndex]
- 785 SessionCnt[FragIndex] += 1
- 786 • DUT must respond with the command FragSessionSetupAns with all bits set to 0. The response
787 must be sent after a delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink.
- 788 • TCL sends fragmented broadcast using multicast ping-slots, multicast address and session
789 context. TCL multicasts the 6 fragments and the 4 redundancy frames until the DUT sends the
790 FragDataBlockReceivedReq.
- 791 • As soon as the full file is received, the DUT must send the FragDataBlockReceivedReq
792 command after a random delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink.
- 793 • TCL responds with the FragDataBlockReceivedAns.
- 794 • TCL reverts the Periodicity to 5 seconds.
- 795 • TCL sends the command FragSessionSetupReq to the DUT using multicast ping slots.
- 796 • DUT must ignore the command as FragSessionSetupReq must be sent only as a unicast.
- 797 • TCL sends the command FragSessionDeleteReq to the DUT using multicast ping slots.
- 798 • DUT must ignore the command as FragSessionDeleteReq must be sent only as a unicast.
- 799 • Wait for Session Time Out.
- 800 • TCL sends the McGroupDeleteReq command to the DUT to request the deletion of the earlier
801 defined group McGroupID = 0.

- The DUT must respond with McGroupDeleteAns command, mentioning no error and the correct McGroupID.
- TCL sends the FragSessionDeleteReq command to the DUT with FragIndex = 0
- DUT must respond with FragSessionDeleteAns command with the bit 2 set to 0 and the FragIndex = 0

Class C - Multicast and Fragmentation Setup

Note: If the device does not support multicast and Class C, skip to the next test.

- TCL creates a multicast session on the DUT using McGroupSetupReq with the following parameters:
 - McGroupID = 0
 - McAddr = 0x00ABCDEF
 - McKey = 0x0102030405060708090A0B0C0D0E0F10
 - minMcFCount = 0
 - maxMcFCount = 0x0000FFFF
- DUT must respond with the McGroupSetupAns command without any errors, with the correct McGroupID.
- TCL sends the McClassCSessionReq command to the DUT to setup a temporary Class C multicast session with
 - McGroupID = 0
 - SessionTime = 30 seconds from the current time
 - SessionTimeOut = 128 seconds
 - DLFreq = Any allowed frequency other than the default RX2 freq (refer to Regional Parameters Specification)
 - DR = Max125KHzDR (refer to Regional Parameters Specification)
- The DUT must respond with McClassCSessionAns command with
 - McGroupID = 0
 - DRError = 0
 - FreqError = 0
 - McGroupUndefined = 0
 - StartMissed = 0
 - Reserved for future errors = 0
 - TimeToStart = number of seconds from the McClassCSessionAns uplink to the beginning of the multicast session
- TCL sends the command FragSessionSetupReq to the DUT with
 - FragSession
 - FragIndex = 0
 - McGroupBitMask = 0000
 - NbFrag = 6

- 843
 - FragSize = 170
- 844
 - Control
 - 845▪ FragAlgo = 0
 - 846▪ BlockAckDelay = 0
 - 847▪ AckReception = 0
- 848
 - Padding = 25
- 849
 - SessionCnt = SessionCnt[FragIndex]
- 850
 - DUT must respond with the command FragSessionSetupAns with all bits set to 0. The response
- 851
 - must be sent after a delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink.
- 852
 - After the SessionTime, TCL sends fragmented broadcast using multicast and session context
- 853
 - during the RXC(3) window. TCL multicasts the 6 fragments and 4 redundancy frames.
- 854
 - DUT must ignore the multicast fragments as only unicast fragments are allowed.
- 855
 - TCL sends the command FragSessionStatusReq in RXC(3) window to the DUT with FragIndex
- 856
 - = 0 and the Participants = 1
- 857
 - FragDataBlockReceivedReq must not be sent by the DUT
- 858
 - DUT must respond with FragSessionStatusAns with NbFragReceived = 0. The response must
- 859
 - be sent after a delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink.
- 860
 - TCL sends the command FragSessionSetupReq to the DUT with
- 861
 - FragSession
 - 862▪ FragIndex = 0
 - 863▪ McGroupBitMask = 0001
- 864
 - NbFrag = 6
- 865
 - FragSize = 170
- 866
 - Control
 - 867▪ FragAlgo = 0
 - 868▪ BlockAckDelay = 0
 - 869▪ AckReception = 1
- 870
 - Padding = 25
- 871
 - SessionCnt = SessionCnt[FragIndex]
- 872
 - SessionCnt[FragIndex] += 1
- 873
 - DUT must respond with the command FragSessionSetupAns with all bits set to 0. The response
- 874
 - must be sent after a delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink.
- 875
 - TCL sends the command FragSessionStatusReq to the DUT with FragIndex = 0 and the
- 876
 - Participants = 0
- 877
 - TCL sends fragmented broadcast using multicast and session context during the RXC(3)
- 878
 - window. TCL multicasts the 6 fragments and 4 redundancy frames until the DUT sends the
- 879
 - FragDataBlockReceivedReq.
- 880
 - As soon as the full file is received, the DUT must send the FragDataBlockReceivedReq
- 881
 - command after a random delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink.
- 882
 - TCL responds with the FragDataBlockReceivedAns.
- 883
 - TCL reverts the Periodicity to 5 seconds.

- TCL sends the command FragSessionSetupReq to the DUT using multicast ping slots.
- DUT must ignore the command as FragSessionSetupReq must be sent only as a unicast.
- TCL sends the command FragSessionDeleteReq to the DUT using multicast ping slots.
- DUT must ignore the command as FragSessionDeleteReq must be sent only as a unicast.
- Wait for Session Time Out.
- TCL sends the McGroupDeleteReq command to the DUT to request the deletion of the earlier defined groups McGroupID = 0.
- The DUT must respond with McGroupDeleteAns command, mentioning no error and the correct McGroupIDs.
- TCL sends the FragSessionDeleteReq command to the DUT with FragIndex = 0
- DUT must respond with FragSessionDeleteAns command with the bit 2 set to 0 and the FragIndex = 0

This test uses a 995-bytes file whose arbitrary content is:

Fragments size = 170

Padding = 25

Number of uncoded fragments = 6

Number of coded fragments = 4 (Redundancy)

Original data block

6A B8 00 4D 25 18 2F 58 65 89 6B AF 34 E0 07 AB 6A 8E 24 33 CC F7 50 B1 DF E4 16 0A 2B E0 19 6B F4
88 B0 50 AF D5 05 BF FC BF 48 C9 1A 72 E8 4B 49 21 05 AD 36 44 7D 0E 92 25 96 B2 1A 6A B1 6A 0D
89 A9 83 F1 96 E6 23 24 CE 65 2A ED 59 BF B9 E1 9F BF 59 45 E4 6D F6 A9 9F 1D F2 73 93 68 3C E6 92
01 9D 53 86 E2 5B E8 9F 04 ED B0 FE 2C 23 EE B2 11 C1 C0 EB B5 20 05 07 07 3F DB 89 8D D7 20 47 95
F7 8F 05 CC 3B CE 63 DC BF 8E 23 0F 1F 0B 1B 3A B6 8F 03 12 F7 91 34 40 BE 32 94 F7 D8 3D 7E 9E D3
28 05 12 7C 9B 91 51 FC 94 61 8C BE AB 44 11 5E A1 36 C0 11 42 CD 31 A3 86 EC 43 11 BB C5 E7 EE 04
3C 9D F2 F2 8E E9 A4 63 7C 9A 8C EC EA 65 F6 2C 20 22 81 05 F2 D3 04 2D 55 21 CE 58 F0 94 E0 D7 E7
75 8B CC 49 7D 99 04 97 6F CE 50 E4 93 2F C9 9C 0E 6B AD EA 00 F9 60 F8 9A D3 93 A0 49 96 BF DB C1
B2 DC 52 AB 73 61 69 66 51 9F 6E F8 AD 33 6D 58 CB E0 E6 A9 45 40 DA 87 CD 92 BB 84 C5 91 77 57
11 60 14 FB 2E CF DF B0 91 29 77 58 39 97 50 EA E8 42 1C 31 7F BA 35 3F D9 6A 9D 3C 1A 84 7A 27 9F
8B A7 25 C0 39 84 C8 06 53 DF D7 89 DD F2 D3 DA 19 A6 B3 9C CC 09 C4 BB 42 42 A1 58 CB 72 C8 FD
4D 24 E6 8A F9 A2 FD 8B 86 23 5B 07 29 BE 08 5D DC B1 B0 30 71 94 FC 34 3F 43 BF 75 0F 82 36 CC 4C
07 97 D7 61 BF 82 8A F5 CD 08 B5 77 F2 38 44 15 6D 1C A2 CD B2 C3 57 D8 6D D2 A0 25 14 05 11 75
1D 07 C0 65 BE 73 73 7A 79 CD 67 E7 09 C5 20 9E 03 89 01 F3 E7 CB E9 25 28 30 9F E7 FC B5 BB E8 66
40 2C 1E CF 25 43 D1 4F FB 44 88 50 E8 5D 6F 83 EF 08 B7 E3 07 85 53 DB 8E B0 73 A0 4A 02 93 4F 84
EA 6D 3F 5F EE EF D7 EB 3A 16 3A 50 2D 9B 69 D0 2F B3 3D 92 59 0F 3A A9 7F 84 2D 92 FE D0 98 F9 E6
98 08 18 11 73 60 F9 2B F8 C4 D2 A1 AB 7A 03 5A 7D BA 78 75 23 03 C1 52 FB 38 56 86 C0 76 20 50 81
AC C4 21 06 84 CF 03 AB AF 73 E9 A4 01 7C DB D4 A6 AC 94 46 8F AB 5A DA 32 BF 4A C5 6D CE 5A 36
C4 4F BB BE 38 37 33 24 60 07 1C AC CC 15 3B 35 EA B5 8D 4E D5 6F EB B4 7A 20 F9 29 34 6E 67 25 BA
30 A4 C0 36 99 BF A3 98 4B BB F1 6D C7 0E D5 31 65 4D 14 E7 5E 87 7E 22 35 13 82 43 5B 1C C9 1B FB

925 2D 92 0B C9 30 87 BD 26 8D 37 C2 B8 2D DC 05 DB 8F 67 C1 B7 FC 47 01 EE DB BA 84 B4 C7 60 C4 BF
926 9C 66 B2 01 C6 E5 3D 1F 38 4D E1 8B 49 23 4A 9D 53 75 71 D3 6D 58 AC 38 77 50 A0 E0 72 C8 75 A7
927 22 6E E8 9A C4 81 7F D7 11 92 F0 84 32 D8 40 B3 8A F2 9F D6 02 FC 14 52 F1 02 D2 DC 70 41 CD 7A 22
928 ED E4 7D DA 6B AE 65 81 30 F6 4B 1A 25 04 B7 90 CB 81 CA B1 C6 68 A5 2E 52 2C 68 3E 68 F9 52 FB
929 A2 60 DB 9E 40 CA 6E 5B 54 B2 45 CE 4B 8B 7C DA E3 2F 95 E5 72 EB 47 9B AE 3A 04 6A EF 57 C7 2D
930 57 25 B7 B2 B0 41 B1 3A 6C 5F 5B 0F A1 B4 9C A5 2B 26 83 DF 2F 76 6D 7F 29 57 43 D7 CC 6D 9B 25
931 82 4C DB AB A2 20 78 FC F2 A4 27 A3 90 78 6D 99 D9 C0 94 EC 11 FD 0E 33 6C 1B 9F 0C 49 10 B3 AA
932 61 30 BE 57 CB 7C 86 07 A4 59 3A 6F 62 78 FA 5D C5 8D E3 5B 3F E8 0B F2 8E 60 FE 0F 84 08 92 2E A1
933 FA DF 73 B5 C6 7E 87 26 5E 24 B9 7A 72 E2 87 68 45 12 6B 07 4A 80 F6 1C AC 7F C6 25 15 66 CB 31 C4
934 4A 37 04 66 61 A8 12 27 04 1D A6 67 52 8E FD D5 B2 EA 0A 12 79 59 EF 7D 8A E4 72 E0 41 46 54 8C 38
935 AB 24 18 DE 3C 62 92 86 13 DF F3 CF 48 87 57 8D

936

937 **Fragment 1**

938 0x6A, 0xB8, 0x00, 0x4D, 0x25, 0x18, 0x2F, 0x58, 0x65, 0x89, 0x6B, 0xAF, 0x34, 0xE0, 0x07, 0xAB,
939 0x6A, 0x8E, 0x24, 0x33, 0xCC, 0xF7, 0x50, 0xB1, 0xDF, 0xE4, 0x16, 0x0A, 0x2B, 0xE0, 0x19, 0x6B,
940 0xF4, 0x88, 0xB0, 0x50, 0xAF, 0xD5, 0x05, 0xBF, 0xFC, 0xBF, 0x48, 0xC9, 0x1A, 0x72, 0xE8, 0x4B,
941 0x49, 0x21, 0x05, 0xAD, 0x36, 0x44, 0x7D, 0x0E, 0x92, 0x25, 0x96, 0xB2, 0x1A, 0x6A, 0xB1, 0x6A,
942 0x0D, 0x89, 0xA9, 0x83, 0xF1, 0x96, 0xE6, 0x23, 0x24, 0xCE, 0x65, 0x2A, 0xED, 0x59, 0xBF, 0xB9,
943 0xE1, 0x9F, 0xBF, 0x59, 0x45, 0xE4, 0x6D, 0xF6, 0xA9, 0x9F, 0x1D, 0xF2, 0x73, 0x93, 0x68, 0x3C,
944 0xE6, 0x92, 0x01, 0x9D, 0x53, 0x86, 0xE2, 0x5B, 0xE8, 0x9F, 0x04, 0xED, 0xB0, 0xFE, 0x2C, 0x23,
945 0xEE, 0xB2, 0x11, 0xC1, 0xC0, 0xEB, 0xB5, 0x20, 0x05, 0x07, 0x07, 0x3F, 0xDB, 0x89, 0x8D, 0xD7,
946 0x20, 0x47, 0x95, 0xF7, 0x8F, 0x05, 0xCC, 0x3B, 0xCE, 0x63, 0xDC, 0xBF, 0x8E, 0x23, 0x0F, 0x1F,
947 0x0B, 0x1B, 0x3A, 0xB6, 0x8F, 0x03, 0x12, 0xF7, 0x91, 0x34, 0x40, 0xBE, 0x32, 0x94, 0xF7, 0xD8,
948 0x3D, 0x7E, 0x9E, 0xD3, 0x28, 0x05, 0x12, 0x7C, 0x9B, 0x91,

949

950 **Fragment 2**

951 0x51, 0xFC, 0x94, 0x61, 0x8C, 0xBE, 0xAB, 0x44, 0x11, 0x5E, 0xA1, 0x36, 0xC0, 0x11, 0x42, 0xCD,
952 0x31, 0xA3, 0x86, 0xEC, 0x43, 0x11, 0xBB, 0xC5, 0xE7, 0xEE, 0x04, 0x3C, 0x9D, 0xF2, 0xF2, 0x8E,
953 0xE9, 0xA4, 0x63, 0x7C, 0x9A, 0x8C, 0xEC, 0xEA, 0x65, 0xF6, 0x2C, 0x20, 0x22, 0x81, 0x05, 0xF2,
954 0xD3, 0x04, 0x2D, 0x55, 0x21, 0xCE, 0x58, 0xF0, 0x94, 0xE0, 0xD7, 0xE7, 0x75, 0x8B, 0xCC, 0x49,
955 0x7D, 0x99, 0x04, 0x97, 0x6F, 0xCE, 0x50, 0xE4, 0x93, 0x2F, 0xC9, 0x9C, 0x0E, 0x6B, 0xAD, 0xEA,
956 0x00, 0xF9, 0x60, 0xF8, 0x9A, 0xD3, 0x93, 0xA0, 0x49, 0x96, 0xBF, 0xDB, 0xC1, 0xB2, 0xDC, 0x52,
957 0xAB, 0x73, 0x61, 0x69, 0x66, 0x51, 0x9F, 0x6E, 0xF8, 0xAD, 0x33, 0x6D, 0x58, 0xCB, 0xE0, 0xE6,
958 0xA9, 0x45, 0x40, 0xDA, 0x87, 0xCD, 0x92, 0xBB, 0x84, 0xC5, 0x91, 0x77, 0x57, 0x11, 0x60, 0x14,
959 0xFB, 0x2E, 0xCF, 0xDF, 0xB0, 0x91, 0x29, 0x77, 0x58, 0x39, 0x97, 0x50, 0xEA, 0xE8, 0x42, 0x1C,
960 0x31, 0x7F, 0xBA, 0x35, 0x3F, 0xD9, 0x6A, 0x9D, 0x3C, 0x1A, 0x84, 0x7A, 0x27, 0x9F, 0x8B, 0xA7,
961 0x25, 0xC0, 0x39, 0x84, 0xC8, 0x06, 0x53, 0xDF, 0xD7, 0x89,

962

963 **Fragment 3**

964 0xDD, 0xF2, 0xD3, 0xDA, 0x19, 0xA6, 0xB3, 0x9C, 0xCC, 0x09, 0xC4, 0xBB, 0x42, 0x42, 0xA1, 0x58,
965 0xCB, 0x72, 0xC8, 0xFD, 0x4D, 0x24, 0xE6, 0x8A, 0xF9, 0xA2, 0xFD, 0x8B, 0x86, 0x23, 0x5B, 0x07,

0x29, 0xBE, 0x08, 0x5D, 0xDC, 0xB1, 0xB0, 0x30, 0x71, 0x94, 0xFC, 0x34, 0x3F, 0x43, 0xBF, 0x75,
0x0F, 0x82, 0x36, 0xCC, 0x4C, 0x07, 0x97, 0xD7, 0x61, 0xBF, 0x82, 0x8A, 0xF5, 0xCD, 0x08, 0xB5,
0x77, 0xF2, 0x38, 0x44, 0x15, 0x6D, 0x1C, 0xA2, 0xCD, 0xB2, 0xC3, 0x57, 0xD8, 0x6D, 0xD2, 0xA0,
0x25, 0x14, 0x05, 0x11, 0x75, 0x1D, 0x07, 0xC0, 0x65, 0xBE, 0x73, 0x73, 0x7A, 0x79, 0xCD, 0x67,
0xE7, 0x09, 0xC5, 0x20, 0x9E, 0x03, 0x89, 0x01, 0xF3, 0xE7, 0xCB, 0xE9, 0x25, 0x28, 0x30, 0x9F,
0xE7, 0xFC, 0xB5, 0xBB, 0xE8, 0x66, 0x40, 0x2C, 0x1E, 0xCF, 0x25, 0x43, 0xD1, 0x4F, 0xFB, 0x44,
0x88, 0x50, 0xE8, 0x5D, 0x6F, 0x83, 0xEF, 0x08, 0xB7, 0xE3, 0x07, 0x85, 0x53, 0xDB, 0x8E, 0xB0,
0x73, 0xA0, 0x4A, 0x02, 0x93, 0x4F, 0x84, 0xEA, 0x6D, 0x3F, 0x5F, 0xEE, 0xEF, 0xD7, 0xEB, 0x3A,
0x16, 0x3A, 0x50, 0x2D, 0x9B, 0x69, 0xD0, 0x2F, 0xB3, 0x3D,

Fragment 4

0x92, 0x59, 0x0F, 0x3A, 0xA9, 0x7F, 0x84, 0x2D, 0x92, 0xFE, 0xD0, 0x98, 0xF9, 0xE6, 0x98, 0x08,
0x18, 0x11, 0x73, 0x60, 0xF9, 0x2B, 0xF8, 0xC4, 0xD2, 0xA1, 0xAB, 0x7A, 0x03, 0x5A, 0x7D, 0xBA,
0x78, 0x75, 0x23, 0x03, 0xC1, 0x52, 0xFB, 0x38, 0x56, 0x86, 0xC0, 0x76, 0x20, 0x50, 0x81, 0xAC,
0xC4, 0x21, 0x06, 0x84, 0xCF, 0x03, 0xAB, 0xAF, 0x73, 0xE9, 0xA4, 0x01, 0x7C, 0xDB, 0xD4, 0xA6,
0xAC, 0x94, 0x46, 0x8F, 0xAB, 0x5A, 0xDA, 0x32, 0xBF, 0x4A, 0xC5, 0x6D, 0xCE, 0x5A, 0x36, 0xC4,
0x4F, 0xBB, 0xBE, 0x38, 0x37, 0x33, 0x24, 0x60, 0x07, 0x1C, 0xAC, 0xCC, 0x15, 0x3B, 0x35, 0xEA,
0xB5, 0x8D, 0x4E, 0xD5, 0x6F, 0xEB, 0xB4, 0x7A, 0x20, 0xF9, 0x29, 0x34, 0x6E, 0x67, 0x25, 0xBA,
0x30, 0xA4, 0xC0, 0x36, 0x99, 0xBF, 0xA3, 0x98, 0x4B, 0xBB, 0xF1, 0x6D, 0xC7, 0x0E, 0xD5, 0x31,
0x65, 0x4D, 0x14, 0xE7, 0x5E, 0x87, 0x7E, 0x22, 0x35, 0x13, 0x82, 0x43, 0x5B, 0x1C, 0xC9, 0x1B,
0xFB, 0x2D, 0x92, 0x0B, 0xC9, 0x30, 0x87, 0xBD, 0x26, 0x8D, 0x37, 0xC2, 0xB8, 0x2D, 0xDC, 0x05,
0xDB, 0x8F, 0x67, 0xC1, 0xB7, 0xFC, 0x47, 0x01, 0xEE, 0xDB,

Fragment 5

0xBA, 0x84, 0xB4, 0xC7, 0x60, 0xC4, 0xBF, 0x9C, 0x66, 0xB2, 0x01, 0xC6, 0xE5, 0x3D, 0x1F, 0x38,
0x4D, 0xE1, 0x8B, 0x49, 0x23, 0x4A, 0x9D, 0x53, 0x75, 0x71, 0xD3, 0x6D, 0x58, 0xAC, 0x38, 0x77,
0x50, 0xA0, 0xE0, 0x72, 0xC8, 0x75, 0xA7, 0x22, 0x6E, 0xE8, 0x9A, 0xC4, 0x81, 0x7F, 0xD7, 0x11,
0x92, 0xF0, 0x84, 0x32, 0xD8, 0x40, 0xB3, 0x8A, 0xF2, 0x9F, 0xD6, 0x02, 0xFC, 0x14, 0x52, 0xF1,
0x02, 0xD2, 0xDC, 0x70, 0x41, 0xCD, 0x7A, 0x22, 0xED, 0xE4, 0x7D, 0xDA, 0x6B, 0xAE, 0x65, 0x81,
0x30, 0xF6, 0x4B, 0x1A, 0x25, 0x04, 0xB7, 0x90, 0xCB, 0x81, 0xCA, 0xB1, 0xC6, 0x68, 0xA5, 0x2E,
0x52, 0x2C, 0x68, 0x3E, 0x68, 0xF9, 0x52, 0xFB, 0xA2, 0x60, 0xDB, 0x9E, 0x40, 0xCA, 0x6E, 0x5B,
0x54, 0xB2, 0x45, 0xCE, 0x4B, 0x8B, 0x7C, 0xDA, 0xE3, 0x2F, 0x95, 0xE5, 0x72, 0xEB, 0x47, 0x9B,
0xAE, 0x3A, 0x04, 0x6A, 0xEF, 0x57, 0xC7, 0x2D, 0x57, 0x25, 0xB7, 0xB2, 0xB0, 0x41, 0xB1, 0x3A,
0x6C, 0x5F, 0x5B, 0x0F, 0xA1, 0xB4, 0x9C, 0xA5, 0x2B, 0x26, 0x83, 0xDF, 0x2F, 0x76, 0x6D, 0x7F,
0x29, 0x57, 0x43, 0xD7, 0xCC, 0x6D, 0x9B, 0x25, 0x82, 0x4C,

Fragment 6

0xDB, 0xAB, 0xA2, 0x20, 0x78, 0xFC, 0xF2, 0xA4, 0x27, 0xA3, 0x90, 0x78, 0x6D, 0x99, 0xD9, 0xC0,
0x94, 0xEC, 0x11, 0xFD, 0x0E, 0x33, 0x6C, 0x1B, 0x9F, 0x0C, 0x49, 0x10, 0xB3, 0xAA, 0x61, 0x30,
0xBE, 0x57, 0xCB, 0x7C, 0x86, 0x07, 0xA4, 0x59, 0x3A, 0x6F, 0x62, 0x78, 0xFA, 0x5D, 0xC5, 0x8D,
0xE3, 0x5B, 0x3F, 0xE8, 0x0B, 0xF2, 0x8E, 0x60, 0xFE, 0x0F, 0x84, 0x08, 0x92, 0x2E, 0xA1, 0xFA,

1007 0xDF, 0x73, 0xB5, 0xC6, 0x7E, 0x87, 0x26, 0x5E, 0x24, 0xB9, 0x7A, 0x72, 0xE2, 0x87, 0x68, 0x45,
1008 0x12, 0x6B, 0x07, 0x4A, 0x80, 0xF6, 0x1C, 0xAC, 0x7F, 0xC6, 0x25, 0x15, 0x66, 0xCB, 0x31, 0xC4,
1009 0x4A, 0x37, 0x04, 0x66, 0x61, 0xA8, 0x12, 0x27, 0x04, 0x1D, 0xA6, 0x67, 0x52, 0x8E, 0xFD, 0xD5,
1010 0xB2, 0xEA, 0x0A, 0x12, 0x79, 0x59, 0xEF, 0x7D, 0x8A, 0xE4, 0x72, 0xE0, 0x41, 0x46, 0x54, 0x8C,
1011 0x38, 0xAB, 0x24, 0x18, 0xDE, 0x3C, 0x62, 0x92, 0x86, 0x13, 0xDF, 0xF3, 0xCF, 0x48, 0x87, 0x57,
1012 0x8D, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
1013 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00,
1014

Fragment 7 (Redundancy Fragment 1)

1016 0x81, 0xC0, 0x20, 0xEB, 0xC9, 0x62, 0x3B, 0x80, 0x12, 0x65, 0xCB, 0x5F, 0x11, 0xCC, 0x5A, 0x5E,
1017 0x16, 0xCC, 0x29, 0x96, 0xAC, 0xAC, 0x76, 0x27, 0x4D, 0x7B, 0xC1, 0x5B, 0xEE, 0xBE, 0xD3, 0x92,
1018 0x4D, 0x8C, 0x33, 0x5E, 0xFD, 0x2C, 0x4E, 0x77, 0xF7, 0xA1, 0xFE, 0x2D, 0xB9, 0x8C, 0x3A, 0xA8,
1019 0x08, 0xD5, 0xAC, 0xCA, 0xCF, 0xCA, 0x96, 0x74, 0xF4, 0x5A, 0x97, 0x57, 0x93, 0xF5, 0x2F, 0xD2,
1020 0x72, 0xC2, 0x71, 0x64, 0xDF, 0x95, 0xCC, 0xE5, 0x5A, 0x05, 0xD1, 0x6C, 0x88, 0x9C, 0x77, 0xD2,
1021 0xD1, 0x90, 0x94, 0xBB, 0xFA, 0x33, 0x49, 0xC6, 0x2B, 0x88, 0x68, 0x98, 0x74, 0x49, 0x11, 0x40,
1022 0x1F, 0xCD, 0x08, 0xCA, 0x5D, 0x2E, 0x2F, 0xCE, 0xB2, 0x52, 0xEC, 0x1E, 0xA8, 0xFF, 0xA2, 0x9E,
1023 0x13, 0x45, 0x14, 0xD5, 0x0C, 0xAD, 0x5B, 0x41, 0x62, 0xED, 0x03, 0xAD, 0xFE, 0x73, 0xAA, 0x58,
1024 0x75, 0x53, 0x5E, 0x42, 0xD0, 0xC3, 0x22, 0x61, 0xC1, 0x7F, 0xFC, 0x5D, 0xD4, 0x8A, 0xFC, 0x39,
1025 0x56, 0x3B, 0xDB, 0x8C, 0x11, 0x6E, 0xE4, 0xCF, 0x86, 0x08, 0x47, 0x1B, 0x3A, 0x7D, 0x11, 0x00,
1026 0x31, 0xE9, 0xE4, 0x80, 0x2C, 0x6E, 0xDA, 0x86, 0xCE, 0x54,
1027

Fragment 8 (Redundancy Fragment 2)

1029 0x42, 0x65, 0xBB, 0xB0, 0xEC, 0xA3, 0x14, 0xE9, 0x91, 0xC5, 0xBA, 0xF1, 0x28, 0x3B, 0x80, 0x9B,
1030 0x3F, 0x7E, 0xDC, 0x1A, 0x16, 0x96, 0x35, 0x26, 0x78, 0x34, 0x6E, 0x1D, 0x70, 0x16, 0x5C, 0xA6,
1031 0xDC, 0x5D, 0x73, 0x21, 0xA6, 0xF2, 0x59, 0xA5, 0xC4, 0xD1, 0x12, 0x7B, 0xBB, 0x5D, 0xBE, 0xF6,
1032 0x1F, 0xF0, 0x87, 0x1B, 0x21, 0x07, 0x65, 0x2B, 0x13, 0x53, 0xE4, 0xB1, 0x9A, 0xA5, 0x37, 0x3D,
1033 0xA3, 0xCF, 0x33, 0x7C, 0x1B, 0x01, 0x46, 0x33, 0x76, 0x60, 0xDD, 0x9D, 0x48, 0xAD, 0xEC, 0xFC,
1034 0x9E, 0xD2, 0x4A, 0x7B, 0x57, 0xD3, 0xFE, 0x06, 0x65, 0x02, 0x7B, 0x8F, 0xA0, 0xC0, 0xF8, 0xF8,
1035 0x01, 0x33, 0x27, 0x76, 0x54, 0x94, 0x04, 0xDA, 0x6A, 0x06, 0xF6, 0x47, 0x9E, 0x53, 0x67, 0xC2,
1036 0x8A, 0xA4, 0x94, 0x39, 0x12, 0xDF, 0x6A, 0x62, 0xAD, 0x93, 0x63, 0xB7, 0x6E, 0x6C, 0x1F, 0x7D,
1037 0xEB, 0x30, 0x85, 0x7A, 0x3E, 0xD5, 0x75, 0x34, 0xAC, 0x55, 0xE9, 0x4E, 0x65, 0x7E, 0x77, 0x3E,
1038 0x9C, 0x69, 0xF3, 0xB2, 0xE7, 0x87, 0x09, 0xEF, 0x9C, 0x9F, 0xF4, 0xA3, 0xA5, 0xCF, 0x46, 0xA2,
1039 0xCF, 0xA6, 0xBA, 0xC5, 0x53, 0x94, 0xCE, 0x58, 0xF7, 0x06,
1040

Fragment 9 (Redundancy Fragment 3)

1042 0xA9, 0x1D, 0x9B, 0x16, 0x00, 0xD9, 0x00, 0x31, 0xE6, 0x29, 0x1A, 0x01, 0x0D, 0x17, 0xDD, 0x6E,
1043 0x43, 0x3C, 0xD1, 0xBF, 0x76, 0xCD, 0x13, 0xB0, 0xEA, 0xAB, 0xB9, 0x4C, 0xB5, 0x48, 0x96, 0x5F,
1044 0x65, 0x59, 0xF0, 0x2F, 0xF4, 0x0B, 0x12, 0x6D, 0xCF, 0xCF, 0xA4, 0x9F, 0x18, 0xA3, 0x6C, 0x15,
1045 0x5E, 0x04, 0x2E, 0x7C, 0xD8, 0x89, 0x8E, 0x51, 0x75, 0x2C, 0xE5, 0x54, 0x13, 0x3A, 0xA9, 0x85,
1046 0xDC, 0x84, 0xEB, 0x9B, 0x35, 0x02, 0x6C, 0xF5, 0x08, 0xAB, 0x69, 0xDB, 0x2D, 0x68, 0x24, 0x97,
1047 0xAE, 0xDD, 0x61, 0x99, 0xE8, 0x04, 0xDA, 0x36, 0xE7, 0x15, 0x0E, 0xE5, 0xA7, 0x1A, 0x81, 0x84,

1048 0xF8, 0x6C, 0x2E, 0x21, 0x5A, 0x3C, 0xC9, 0x4F, 0x30, 0xCB, 0x1E, 0xB4, 0x86, 0x52, 0xE9, 0x7F,
1049 0x77, 0x53, 0x91, 0x2D, 0xDE, 0x99, 0x84, 0x03, 0xCA, 0x79, 0x67, 0x25, 0x4B, 0x96, 0x38, 0xF2,
1050 0xBE, 0x24, 0x4E, 0xCF, 0x61, 0x13, 0x9B, 0x6E, 0xA3, 0x49, 0xC9, 0xAC, 0x3F, 0xD7, 0x84, 0x18,
1051 0xC1, 0x49, 0x12, 0x88, 0x79, 0xEA, 0xFF, 0xD7, 0x8B, 0xA3, 0xF3, 0x06, 0xAD, 0x26, 0xA0, 0x7A,
1052 0xC3, 0x31, 0xC0, 0x96, 0x57, 0xFF, 0x06, 0xA2, 0xA2, 0xC3,

1053

1054 **Fragment 10 (Redundancy Fragment 4)**

1055 0xBC, 0xDD, 0xC5, 0x3D, 0x01, 0x9E, 0xFE, 0xA4, 0x8D, 0x18, 0x55, 0x05, 0xCA, 0xE6, 0x67, 0xA0,
1056 0x12, 0x7F, 0x52, 0x49, 0x60, 0x5D, 0x17, 0xC2, 0x13, 0xDF, 0x67, 0xF6, 0x6D, 0x25, 0x02, 0x40,
1057 0xC7, 0x49, 0x23, 0x53, 0x92, 0xC3, 0xB3, 0x4B, 0x25, 0x13, 0x04, 0x88, 0x44, 0x61, 0xAD, 0xE9,
1058 0x7E, 0x29, 0x8D, 0x16, 0x9F, 0xB5, 0xAA, 0x3D, 0x6D, 0x2F, 0xD0, 0x80, 0x9B, 0xF7, 0xFB, 0xBE,
1059 0xAA, 0x53, 0x51, 0xF2, 0x2A, 0x27, 0x40, 0xDE, 0x04, 0xEF, 0xC4, 0xFF, 0x51, 0x44, 0xDF, 0x64,
1060 0x07, 0x89, 0x49, 0x41, 0xD0, 0xEF, 0xAC, 0xFC, 0xD1, 0xF9, 0x9C, 0xD7, 0xDA, 0xDA, 0x59, 0x8D,
1061 0xFF, 0x12, 0xA9, 0x78, 0x97, 0x52, 0xC9, 0xDD, 0x55, 0x9A, 0xB6, 0x10, 0x37, 0x6C, 0xA3, 0x11,
1062 0x01, 0xA4, 0xFA, 0x67, 0xDA, 0xB4, 0xD3, 0x8B, 0x77, 0x04, 0xC2, 0x46, 0xE2, 0xE2, 0xE8, 0x53,
1063 0x1E, 0xC1, 0xC8, 0x2F, 0x5E, 0xE8, 0x4A, 0xB7, 0x66, 0xD5, 0x6F, 0xC4, 0x2C, 0xD2, 0xB8, 0xDD,
1064 0x92, 0xFF, 0x11, 0x0D, 0x32, 0xFB, 0x18, 0x4F, 0x46, 0x19, 0xDC, 0x31, 0xC0, 0xA1, 0x86, 0x45,
1065 0x3F, 0x6D, 0x13, 0xFA, 0x57, 0x04, 0x4B, 0x0A, 0x31, 0x71,

1066

1067 The resulting MIC = 0xAFEEC794 (for SessionCnt[FragIndex = 0] = 0)

1068

1069

1070 **Test Procedure Frame Sequence Chart**

1071

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD PackageVersionReq Payload = [0x]00	
2	DUT sends Unconfirmed frame	→	CMD PackageVersionAns Package Identifier = 3 Package Version = 2 Payload = [0x]00XXXX	
	The TCL sends Unconfirmed frame	←	CP-CMD FragSessionCntReq FragIndex = 0 Payload = [0x]5200	
3	DUT sends Unconfirmed frame	→	CP-CMD FragSessionCntAns SessionCntPrev = XXXX SessionCnt[FragIndex]= SessionCntPrev + 1 Payload = [0x]5200XXXX	
	The TCL sends Unconfirmed frame	←	CMD FragSessionSetupReq FragSession FragIndex = 0 McGroupBitMask = 0000 NbFrag = 6 FragSize = 170 FragAlgo = 0 BlockAckDelay = 0 AckReception = 1 Padding = 25 Descriptor = obtained from the Certification Questionnaire SessionCnt = SessionCnt[FragIndex] MIC = 0XXXXXXXXX (depends on SessionCnt) Payload = [0x]02 XX(for 16 octets) SessionCnt[FragIndex] += 1	
4	DUT sends Unconfirmed frame	→	CMD FragSessionSetupAns Payload = [0x]0200	
5	DUT sends Unconfirmed frame	→ R[5]		
	The TCL sends Unconfirmed frame Send only fragment 2, 3, 4, 5 and 6. Skip fragment 1	← R[5]	From a unicast address CMD DataFragment FragIndex = as set above Payload = [0x]08 XX...	
6	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD FragSessionStatusReq Participants = 0 FragIndex = 0 Payload = [0x]01XX	

7	DUT sends Unconfirmed frame	→	CMD FragSessionStatusAns Status = 0 FragIndex = 0 NbFragReceived = 5 (or 4 if there is a missing fragment) MissingFrag = 1 (or 2 if there is a missing fragment) Payload = [0x]01XXXXXXXX	
8	DUT sends Unconfirmed frame	→ R [max 4]		
	The TCL sends Unconfirmed frame Repeat for a max of 4 redundancy fragments until the DUT sends the FragDataBlockReceivedReq	← R [max 4]	From a unicast address CMD DataFragment FragIndex = as set above Payload = [0x]08 XX...	
9	Once the data block is fully received, DUT sends Unconfirmed frame	→	CMD FragDataBlockReceivedReq MICErr = 0 Payload = [0x]04XX	
	The TCL sends Unconfirmed frame	←	CMD FragDataBlockReceivedAns FragIndex = 0 Payload = [0x]04XX	
10	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD FragSessionStatusReq Participants = 0 FragIndex = 0 Payload = [0x]01XX	
11	DUT sends Unconfirmed frame	→		DUT does not respond to FragSessionStatusReq as full data is received. LCTT waits for ~16 sec (3 uplinks) to check this as BlockAckDelay = 0

	The TCL sends Unconfirmed frame	←	CMD FragSessionSetupReq FragSession FragIndex = 0 McGroupBitMask = 0000 NbFrag = 6 FragSize = 170 FragAlgo = 0 BlockAckDelay = 0 AckReception = 1 Padding = 25 Descriptor = obtained from the Certification Questionnaire SessionCnt = SessionCnt[FragIndex] MIC = 0x12345678 Payload = [0x]02 XX(for 16 octets) SessionCnt[FragIndex] += 1	Increment SessionCnt and invalid MIC check
12	DUT sends Unconfirmed frame	→	CMD FragSessionSetupAns Payload = [0x]0200	
13	DUT sends Unconfirmed frame	→ R[6]		
	The TCL sends Unconfirmed frame	← R[6]	From a unicast address CMD DataFragment FragIndex = as set above Payload = [0x]08 XX...	
	Repeat for all 6 fragments			
14	DUT sends Unconfirmed frame	→ R [4]		
	If FragDataBlockReceivedReq is not sent, the TCL sends Unconfirmed frame If FragDataBlockReceivedReq is sent, skip to next step. Repeat for 4 redundancy frames	← R [4]	From a unicast address CMD DataFragment FragIndex = as set above Payload = [0x]08 XX...	
15	Once the data block is fully received, DUT sends Unconfirmed frame	→	CMD FragDataBlockReceivedReq MICErr = 1 Payload = [0x]04XX	
	The TCL sends Unconfirmed frame	←	CMD FragDataBlockReceivedAns FragIndex = 0 Payload = [0x]04XX	
16	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD FragSessionDeleteReq FragIndex = 0 Payload = [0x]0300	
17	DUT sends Unconfirmed frame	→	CMD FragSessionDeleteAns Bit 2 = 0 FragIndex = 0 Payload = [0x]0300	

	The TCL sends Unconfirmed frame	←	CMD FragSessionSetupReq FragSession FragIndex = 0 McGroupBitMask = 0000 NbFrag = 6 FragSize = 170 FragAlgo = 0 BlockAckDelay = 0 AckReception = 1 Padding = 25 Descriptor = obtained from the Certification Questionnaire MIC = 0XXXXXXXXX (depends on SessionCnt) SessionCnt = PrevSessionCnt (incorrect SessionCnt) Payload = [0x]02 XX(for 16 octets)	Incorrect SessionCnt
18	DUT sends Unconfirmed frame	→	CMD FragSessionSetupAns SessionCnt replay = 1 Payload = [0x]02XX The response must be sent after a delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink.	
	The TCL sends Unconfirmed frame	←	CMD FragSessionDeleteReq FragIndex = 0 Payload = [0x]03XX	
19	DUT sends Unconfirmed frame	→	CMD FragSessionDeleteAns Bit 2 = 1 FragIndex = 0 Payload = [0x]03XX	
	The TCL sends Unconfirmed frame	←	CMD FragSessionStatusReq Participants = 1 FragIndex = 0 Payload = [0x]01XX	
20	DUT sends Unconfirmed frame	→	CMD FragSessionStatusAns Bit 2 of Status field = 1 Received&index not present MissingFrag not present Payload = [0x]0104	
	Perform the next section of tests only if the device supports multicast and Class B Note: It is assumed the end-device has successfully performed beacon acquisition before the temporary Class B multicast session starts			
21	DUT sends Unconfirmed frame	→		
	TCL sends Unconfirmed frame	←	CP-CMD DeviceTimeReq FPort = 224 Payload = [0x]21	For device clock sync
22	DUT sends Unconfirmed frame	→	MAC-CMD DeviceTimeReq Payload = [0x]0D	

	The TCL sends Unconfirmed frame	←	MAC-CMD DeviceTimeAns Payload = [0x]0DXXXXXXXXXX	
23	TCL continuously sends beacons	← R	BCNPayload	
24	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD McGroupSetupReq McGroupID = 0 McAddr = 0x00ABCDEF McKey_encrypted = aes128_decrypt (McKEKey, McKey), where McKey = 0x0102030405060708090A0B0C0D 0E0F10 minMcFCount = 0 maxMcFCount = 0x0000FFFF Payload = [0x]02 XX(for 29 octets)	
25	DUT sends Unconfirmed frame	→	CMD McGroupSetupAns IDerror = 0 McGroupID = 0 Payload = [0x]0200	
	The TCL sends Unconfirmed frame	←	CMD McClassBSessionReq McGroupID = 0 SessionTime = 3*128 seconds from current time Periodicity = 0 TimeOut = 128 seconds DLFreq = Any allowed frequency, refer [2] DR = Max125KHzDR-1 or Max500kHzDR-1, refer [2] Payload = [0x]05 XX(for 10 octets)	
26	DUT sends Unconfirmed frame	→	CMD McClassBSessionAns McGroupID = 0 DRError = 0 FreqError = 0 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 TimeToStart = the number of seconds from the McClassBSessionAns uplink to the beginning of the multicast session Payload = [0x]05XXXXXXXXXX	

	The TCL sends Unconfirmed frame	←	CMD FragSessionSetupReq FragSession FragIndex = 0 McGroupBitMask = 0000 NbFrag = 6 FragSize = 170 FragAlgo = 0 BlockAckDelay = 0 AckReception = 1 Padding = 25 Descriptor = obtained from the Certification Questionnaire MIC = 0XXXXXXXXX (depends on SessionCnt) SessionCnt = SessionCnt[FragIndex] Payload = [0x]02 XX(for 16 octets)	
27	DUT sends Unconfirmed frame Note: Other uplinks may be sent by the DUT at any time while waiting for the condition to be fulfilled.	→	CMD FragSessionSetupAns Payload = [0x]02XX The response must be sent after a delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink.	
	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 30s Payload = [0x]0604	Uplink Periodicity set to 30s
	Wait for SessionTime			
28	DUT sends Unconfirmed frame	→		
29	After the session time, the TCL sends Unconfirmed frame using multicast ping-slots, multicast address and session context Repeat for 6 fragments	← R [6]	CMD DataFragment FragIndex = 0 Payload = [0x]08 XX...	Multicast is not allowed
30	TCL sends Unconfirmed frame using multicast ping-slots, multicast address and session context Repeat for 4 redundancy fragments	← R [4]	CMD DataFragment FragIndex = 0 Payload = [0x]08 XX...	These fragments are ignored by the DUT
31	TCL sends Unconfirmed frame using multicast ping-slots, multicast address and session context	←	CMD FragSessionStatusReq Participants = 1 FragIndex = 0 Payload = [0x]01XX	

32	DUT sends Unconfirmed frame	→	FragDataBlockReceivedReq is not sent by the DUT CMD FragSessionStatusAns Status = 0 FragIndex = 0 NbFragReceived = 0 Payload = [0x]01XXXXXXXX The response must be sent after a delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink	
	The TCL sends Unconfirmed frame	←	CMD FragSessionSetupReq FragSession FragIndex = 0 McGroupBitMask = 0001 NbFrag = 6 FragSize = 170 FragAlgo = 0 BlockAckDelay = 0 AckReception = 1 Padding = 25 Descriptor = obtained from the Certification Questionnaire MIC = 0XXXXXXXX (depends on SessionCnt) SessionCnt = SessionCnt[FragIndex] Payload = [0x]02 XX(for 16 octets) SessionCnt[FragIndex] += 1	
33	DUT sends Unconfirmed frame	→	CMD FragSessionSetupAns Payload = [0x]02XX The response must be sent after a delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink.	
34	TCL sends Unconfirmed frame using multicast ping-slots, multicast address and session context	← R [6]	CMD DataFragment FragIndex = 0 Payload = [0x]08 XX...	
	Repeat for 6 fragments			
35	If FragDataBlockReceivedReq is not sent, the TCL sends Unconfirmed frame using multicast ping-slots, multicast address and session context If FragDataBlockReceivedReq is sent, skip to next step. Repeat for 4 redundancy fragments	← R [4]	CMD DataFragment FragIndex = 0 Payload = [0x]08 XX...	

36	Once the data block is fully received, DUT sends Unconfirmed frame	→	FragDataBlockReceivedReq must be sent after a random delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink CMD FragDataBlockReceivedReq MICErr = 0 Payload = [0x]04XX	
	The TCL sends Unconfirmed frame	←	CMD FragDataBlockReceivedAns FragIndex = 0 Payload = [0x]04XX	
37	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 5s Payload = [0x]0601	Uplink Periodicity set to 5s
38	The TCL sends Unconfirmed frame using multicast ping-slots, multicast address and session context	←	CMD FragSessionSetupReq FragSession FragIndex = 0 McGroupBitMask = 0001 NbFrag = 6 FragSize = 170 FragAlgo = 0 BlockAckDelay = 0 AckReception = 1 Padding = 25 Descriptor = obtained from the Certification Questionnaire MIC = 0XXXXXXXXX (depends on SessionCnt) SessionCnt = SessionCnt[FragIndex] Payload = [0x]02 XX(for 16 octets)	
39	DUT sends Unconfirmed frame	→	FragSessionSetupAns is not sent	FragSessionSetupReq must be sent only as a unicast
40	The TCL sends Unconfirmed frame using multicast ping-slots, multicast address and session context	←	CMD FragSessionDeleteReq FragIndex = 0 Payload = [0x]03XX	
41	DUT sends Unconfirmed frame	→	FragSessionDeleteAns is not sent	FragSessionDeleteReq must be sent only as a unicast
42	Wait for Session Time Out			
43	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD McGroupDeleteReq McGroupID = 0 Payload = [0x]0300	

44	DUT sends Unconfirmed frame	→	CMD McGroupDeleteAns McGroupUndefined = 0 McGroupID = 0 Payload = [0x]0300	
	The TCL sends Unconfirmed frame	←	CMD FragSessionDeleteReq FragIndex = 0 Payload = [0x]03XX	
45	DUT sends Unconfirmed frame	→	CMD FragSessionDeleteAns Bit 2 = 0 FragIndex = 0 Payload = [0x]03XX	
Perform the next section of tests only if the device supports multicast and Class C				
46	DUT sends Unconfirmed frame	→		
	TCL sends Unconfirmed frame	←	CP-CMD DeviceTimeReq FPort = 224 Payload = [0x]21	For device clock sync
47	DUT sends Unconfirmed frame	→	MAC-CMD DeviceTimeReq Payload = [0x]0D	
	The TCL sends Unconfirmed frame	←	MAC-CMD DeviceTimeAns Payload = [0x]0DXXXXXXXXXX	
48	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD McGroupSetupReq McGroupID = 0 McAddr = 0x00ABCDEF McKey_encrypted = aes128_decrypt (McKEKey, McKey), where McKey = 0x0102030405060708090A0B0C0D 0E0F10 minMcFCount = 0 maxMcFCount = 0x0000FFFF Payload = [0x]02 XX(for 29 octets)	
49	DUT sends Unconfirmed frame	→	CMD McGroupSetupAns IDerror = 0 McGroupID = 0 Payload = [0x]0200	
	The TCL sends Unconfirmed frame	←	CMD McClassCSessionReq McGroupID = 0 SessionTime = 30 seconds from the current time SessionTimeOut = 128 seconds DLFreq = Any allowed frequency other than the default RX2 frequency, refer [2] DR = Max125KHzDR or Max500kHzDR, refer [2] Payload = [0x]04 XX(for 10 octets)	

50	DUT sends Unconfirmed frame	→	CMD McClassCSessionAns McGroupID = 0 DRError = 0 FreqError = 0 McGroupUndefined = 0 StartMissed = 0 Reserved for future errors = 0 TimeToStart = number of seconds from the McClassCSessionAns uplink to the beginning of the multicast session Payload = [0x]04XXXXXXXX	
	The TCL sends Unconfirmed frame	←	CMD FragSessionSetupReq FragSession FragIndex = 0 McGroupBitMask = 0000 NbFrag = 6 FragSize = 170 FragAlgo = 0 BlockAckDelay = 0 AckReception = 1 Padding = 25 SessionCnt = SessionCnt[FragIndex] Descriptor = obtained from the Certification Questionnaire MIC = 0XXXXXXXX (depends on SessionCnt) Payload = [0x]02 XX(for 16 octets)	
51	DUT sends Unconfirmed frame	→	CMD FragSessionSetupAns Payload = [0x]02XX The response must be sent after a delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink.	
	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 30s Payload = [0x]0604	Uplink Periodicity set to 30s
52	After the session time, the TCL sends Unconfirmed frame using multicast and session context during the RXC (3) window 500ms after the gateway completes the transmission of the previous downlink. i.e. Downlink TimeonAir + 500ms Repeat for 6 fragments	← R[6]	CMD DataFragment FragIndex = 0 Payload = [0x]08 XX...	Multicast is not allowed

53	TCL sends Unconfirmed frame using multicast and session context during the RXC (3) window 500ms after the gateway completes the transmission of the previous downlink. i.e. Downlink TimeonAir + 500ms Repeat for 4 redundancy fragments	← R [4]	CMD DataFragment FragIndex = 0 Payload = [0x]08 XX...	These fragments are ignored by the DUT
54	TCL sends Unconfirmed frame using multicast and session context during the RXC (3) window 500ms after the gateway completes the transmission of the previous downlink. i.e. Downlink TimeonAir + 500ms	←	CMD FragSessionStatusReq Participants = 1 FragIndex = 0 Payload = [0x]01XX	
55	DUT sends Unconfirmed frame	→	FragDataBlockReceivedReq must not be sent by the DUT CMD FragSessionStatusAns Status = 0 FragIndex = 0 NbFragReceived = 0 Payload = [0x]01XXXXXXXXX The response must be sent after a delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink	
	The TCL sends Unconfirmed frame	←	CMD FragSessionSetupReq FragSession FragIndex = 0 McGroupBitMask = 0001 NbFrag = 6 FragSize = 170 FragAlgo = 0 BlockAckDelay = 0 AckReception = 1 Padding = 25 Descriptor = obtained from the Certification Questionnaire MIC = 0XXXXXXXXX (depends on SessionCnt) SessionCnt = SessionCnt[FragIndex] Payload = [0x]02 XX(for 16 octets) SessionCnt[FragIndex] += 1	
56	DUT sends Unconfirmed frame	→	CMD FragSessionSetupAns Payload = [0x]02XX The response must be sent after a delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink.	

57	TCL sends Unconfirmed frame using multicast and session context during the RXC (3) window 3s after the uplink is received and for the repeat downlinks, 500ms after the gateway completes the transmission of the previous downlink. i.e. Downlink TimeonAir + 500ms Repeat for 6 fragments	← R[6]	CMD DataFragment FragIndex = 0 Payload = [0x]08 XX...	
58	If FragDataBlockReceivedReq is not sent, TCL sends Unconfirmed frame using multicast and session context during the RXC (3) window 500ms after the gateway completes the transmission of the previous downlink. i.e. Downlink TimeonAir + 500ms If FragDataBlockReceivedReq is sent, skip to the next step. Repeat for 4 redundancy fragments	← R [4]	CMD DataFragment FragIndex = 0 Payload = [0x]08 XX...	
59	Once the data block is fully received, DUT sends Unconfirmed frame	→	FragDataBlockReceivedReq must be sent after a random delay of $\text{rand}().2^{\text{BlockAckDelay}+4}$ seconds after the downlink CMD FragDataBlockReceivedReq MICErrror = 0 Payload = [0x]04XX	
	The TCL sends Unconfirmed frame	←	CMD FragDataBlockReceivedAns FragIndex = 0 Payload = [0x]04XX	
60	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CP-CMD TxPeriodicityChangeReq FPort = 224 Periodicity = 5s Payload = [0x]0601	Uplink Periodicity set to 5s

61	TCL sends Unconfirmed frame using multicast and session context during the RXC (3) window 500ms after the gateway completes the transmission of the previous downlink. i.e. Downlink TimeonAir + 500ms	←	CMD FragSessionSetupReq FragSession FragIndex = 0 McGroupBitMask = 0001 NbFrag = 6 FragSize = 170 FragAlgo = 0 BlockAckDelay = 0 AckReception = 1 Padding = 25 Descriptor = obtained from the Certification Questionnaire MIC = 0XXXXXXXXX (depends on SessionCnt) SessionCnt = SessionCnt[FragIndex] Payload = [0x]02 XX(for 16 octets)	
62	DUT sends Unconfirmed frame	→	FragSessionSetupAns is not sent	FragSession SetupReq must be sent only as a unicast
63	TCL sends Unconfirmed frame using multicast and session context during the RXC (3) window 3s after the uplink is received	←	CMD FragSessionDeleteReq FragIndex = 0 Payload = [0x]03XX	
64	DUT sends Unconfirmed frame	→	FragSessionDeleteAns is not sent	FragSession DeleteReq must be sent only as a unicast
	Wait for Session Time Out			
65	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD McGroupDeleteReq McGroupID = 0 Payload = [0x]0300	
66	DUT sends Unconfirmed frame	→	CMD McGroupDeleteAns McGroupUndefined = 0 McGroupID = 0 Payload = [0x]0300	
	The TCL sends Unconfirmed frame	←	CMD FragSessionDeleteReq FragIndex = 0 Payload = [0x]03XX	
67	DUT sends Unconfirmed frame	→	CMD FragSessionDeleteAns Bit 2 = 0 FragIndex = 0 Payload = [0x]03XX	

1072

6. Firmware Management Protocol

Test Purpose: This test verifies all the Firmware Management Protocol commands from TS006-Firmware-Management-Protocol-v1.0.0 [6].

Note: It is recommended that a default FPort value of 203 be used to promote interoperability. The default FPort value is obtained from the Certification Questionnaire.

PackageVersion

- TCL sends the command PackageVersionReq to the DUT
- DUT must respond with the command PackageVersionAns with Package Identifier = 4 and Package Version = 1

Device Version

- TCL sends the command DevVersionReq to the DUT
- DUT must respond with the command DevVersionAns with the FW and HW versions

Device Reboot Time

- TCL sends the command DevRebootTimeReq to the DUT with the RebootTime = 0x00000000
- DUT must reboot as soon as possible and must not answer with the DevRebootTimeAns.
- TCL triggers DUT to send the DeviceTimeReq.
- The DUT sends the DeviceTimeReq to the TCL. TCL must respond with the DeviceTimeAns.
- TCL sends the command DevRebootTimeReq to the DUT with the RebootTime = 30 seconds from the current time.
- DUT must respond with the command DevRebootTimeAns with the RebootTime indicating the number of seconds between sending the message until reboot.
- The DUT must reboot after 30 seconds.
- TCL triggers DUT to send the DeviceTimeReq.
- The DUT sends the DeviceTimeReq to the TCL. TCL must respond with the DeviceTimeAns.
- TCL sends the command DevRebootTimeReq to the DUT with the RebootTime = 40 seconds from the current time.
- DUT must respond with the command DevRebootTimeAns with the RebootTime indicating the number of seconds between sending the message until reboot.
- Before the DUT reboots, TCL sends the command DevRebootTimeReq to the DUT with the RebootTime = 0xFFFFFFFF.
- DUT must cancel the reboot and respond with the command DevRebootTimeAns with the RebootTime = 0xFFFFFFFF.

1110

1111 • TCL sends the command DevRebootTimeReq to the DUT with the RebootTime = invalid time
1112 (before the current time).

1113 • The DUT may (optional) send the DeviceTimeReq to the TCL. TCL must respond with the
1114 DeviceTimeAns.

1115 • DUT must respond with the command DevRebootTimeAns with the RebootTime =
1116 0x00000000. The DUT must not reboot.

1117

1118 Device Reboot Countdown

1119 • TCL sends the command DevRebootCountdownReq to the DUT with the Countdown =
1120 0x00000000

1121 • DUT must reboot as soon as possible and must not answer with the
1122 DevRebootCountdownAns.

1123

1124 • TCL sends the command DevRebootCountdownReq to the DUT with the Countdown = 30
1125 seconds.

1126 • DUT must respond with the command DevRebootCountdownAns with the Countdown
1127 indicating the number of seconds between sending the DevRebootCountdownAns until
1128 reboot.

1129 • The DUT must reboot in 30 seconds as indicated.

1130

1131 • TCL sends the command DevRebootCountdownReq to the DUT with the Countdown = 40
1132 seconds.

1133 • DUT must respond with the command DevRebootCountdownAns with the Countdown
1134 indicating the number of seconds between sending the DevRebootCountdownAns until
1135 reboot.

1136 • Before the DUT reboots, TCL sends the command DevRebootCountdownReq to the DUT with
1137 the Countdown = 0xFFFFFFFF.

1138 • DUT must cancel the reboot and respond with the command DevRebootCountdownAns with
1139 the Countdown = 0xFFFFFFFF.

1140

1141 • TCL sends the command DevRebootCountdownReq to the DUT with the Countdown = invalid
1142 time (DUT cannot reboot at this time).

1143 • DUT must respond with the command DevRebootCountdownAns with the Countdown =
1144 0x00000000. The DUT must not reboot.

1145

1146 Device Upgrade and Delete Image

1147 • TCL sends the command DevUpgradeImageReq to the DUT.

- 1148
- 1149
- 1150
- 1151
- 1152
- 1153
- 1154
- 1155
- 1156
- DUT must respond with the DevUpgradelImageAns with UpImageStatus = any value (0 – 3) and the nextFirmwareVersion = X (if the firmware upgrade image is present).
 - TCL sends the command DevDeletelImageReq to the DUT with the FirmwareToDeleteVersion = X. X = firmware upgrade image version obtained from the DevUpgradelImageAns above. If the firmware upgrade image is not present, X = any value.
 - DUT must respond with the DevUpgradelImageAns with ErrorNoValidImage = 1 (if there is no image present) or 0 (if an image is present) any valid value, and ErrorInvalidVersion = any valid value.

1157 Test Procedure Frame Sequence Chart

1158

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD PackageVersionReq Payload = [0x]00	
2	DUT sends Unconfirmed frame	→	CMD PackageVersionAns Package Identifier = 4 Package Version = 1 Payload = [0x]00XXXX	
	The TCL sends Unconfirmed frame	←	CMD DevVersionReq Payload = [0x]01	
3	DUT sends Unconfirmed frame	→	CMD DevVersionAns Fw version = Fw version of DUT Hw version = Hw version of DUT Payload = [0x]01XXXXXXXXXXXXXXXXXX	
	The TCL sends Unconfirmed frame	←	CMD DevRebootTimeReq RebootTime = 0x00000000 Payload = [0x]0200000000	
4	DUT must reboot as soon as possible and must not answer with the DevRebootTimeAns.	→		
5	If DUT = OTAA device, DUT MAY send a Join-Request frame If DUT = ABP device, skip this step.	→		
	If the DUT sends a Join-Request in the previous step, TCL sends a Join-Accept frame	←		
6	If the Join procedure was performed, all Section 2 – Pre-Requisite Setup Tests must be performed			
7	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CP-CMD DeviceTimeReq FPort = 224 Payload = [0x]21	
8	DUT sends Unconfirmed frame	→	MAC-CMD DeviceTimeReq Payload = [0x]0D	
	TCL sends Unconfirmed frame	←	MAC-CMD DeviceTimeAns Payload = [0x]0DXXXXXXXXXX	
9	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD DevRebootTimeReq RebootTime = 30 seconds from the current time Payload = [0x]02XXXXXXXX	
10	DUT sends Unconfirmed frame	→	CMD DevRebootTimeAns RebootTime = number of seconds between sending the message until reboot Payload = [0x]02XXXXXXXX	

11	DUT must reboot after 30 seconds			
12	If DUT = OTAA device, DUT MAY send a Join-Request frame	→		
	If DUT = ABP device, skip this step. If the DUT sends a Join-Request in the previous step, TCL sends a Join-Accept frame	←		
13	If the Join procedure was performed, all Section 2 – Pre-Requisite Setup Tests must be performed			
14	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CP-CMD DeviceTimeReq FPort = 224 Payload = [0x]21	
15	DUT sends Unconfirmed frame	→	MAC-CMD DeviceTimeReq Payload = [0x]0D	
	TCL sends Unconfirmed frame	←	MAC-CMD DeviceTimeAns Payload = [0x]0DXXXXXXXXXX	
16	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD DevRebootTimeReq RebootTime = 40 seconds from the current time Payload = [0x]02XXXXXXXXXX	
17	DUT sends Unconfirmed frame	→	CMD DevRebootTimeAns RebootTime = number of seconds between sending the message until reboot Payload = [0x]02XXXXXXXXXX	
	Before the DUT reboots, TCL sends Unconfirmed frame	←	CMD DevRebootTimeReq RebootTime = 0xFFFFFFFF Payload = [0x]02XXXXXXXXXX	
18	DUT sends Unconfirmed frame	→	DUT must not reboot CMD DevRebootTimeAns RebootTime = 0xFFFFFFFF Payload = [0x]02XXXXXXXXXX	
	TCL sends Unconfirmed frame	←	CMD DevRebootTimeReq RebootTime = before the current time (invalid) Payload = [0x]02XXXXXXXXXX	
19	[Optional] DUT sends Unconfirmed frame	→	MAC-CMD DeviceTimeReq Payload = [0x]0D	
	If DeviceTimeReq was sent, the TCL sends Unconfirmed frame	←	MACCMD DeviceTimeAns Payload = [0x]0DXXXXXXXXXX	

20	DUT sends Unconfirmed frame	→	CMD DevRebootTimeAns RebootTime = 0x00000000 Payload = [0x]0200000000 DUT must not reboot	
	The TCL sends Unconfirmed frame	←	CMD DevRebootCountdownReq Countdown = 0x00000000 Payload = [0x]03XXXXXX	
21	DUT must reboot as soon as possible and must not answer with the DevRebootCountdownAns			
22	If DUT = OTAA device, DUT MAY send a Join-Request frame If DUT = ABP device, skip this step.	→		
	If the DUT sends a Join-Request in the previous step, TCL sends a Join-Accept frame	←		
23	If the Join procedure was performed, all Section 2 – Pre-Requisite Setup Tests must be performed			
24	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD DevRebootCountdownReq Countdown = 30 seconds Payload = [0x]03XXXXXX	
25	DUT sends Unconfirmed frame	→	CMD DevRebootCountdownAns Countdown = number of seconds between sending the DevRebootCountdownAns until reboot Payload = [0x]03XXXXXX	
26	DUT must reboot in 30 seconds			
27	If DUT = OTAA device, DUT MAY send a Join-Request frame If DUT = ABP device, skip this step.	→		
	If the DUT sends a Join-Request in the previous step, TCL sends a Join-Accept frame	←		
28	If the Join procedure was performed, all Section 2 – Pre-Requisite Setup Tests must be performed			
29	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD DevRebootCountdownReq Countdown = 40 seconds Payload = [0x]03XXXXXX	

30	DUT sends Unconfirmed frame	→	CMD DevRebootCountdownAns Countdown = number of seconds between sending the DevRebootCountdownAns until reboot Payload = [0x]03XXXXXX	
	Before the DUT reboots, TCL sends Unconfirmed frame	←	CMD DevRebootCountdownReq Countdown = 0FFFFFFF Payload = [0x]03FFFFFF	
31	DUT sends Unconfirmed frame	→	DUT must not reboot CMD DevRebootCountdownAns Countdown = 0FFFFFFF Payload = [0x]03FFFFFF	
Firmware Upgrade tests				
32	DUT sends Unconfirmed frame	→		
	TCL sends Unconfirmed frame	←	CMD DevUpgradelImageReq Payload = [0x]04	
33	DUT sends Unconfirmed frame	→	CMD DevUpgradelImageAns UplImageStatus = any value (0-3) nextFirmwareVersion = X (if the firmware upgrade image is present) Payload = [0x]04XX (cond)XX 4 octets	
	TCL sends Unconfirmed frame	←	CMD DevDeletelImageReq FirmwareToDeleteVersion = X (where X = firmware upgrade image version obtained from the DevUpgradelImageAns above. If the firmware upgrade image is not present, X = any value) Payload = [0x]05XXXXXXXXXX	
34	DUT sends Unconfirmed frame	→	CMD DevDeletelImageAns ErrorNoValidImage - - ErrorNoValidImage = 1 (if there is no image present) or - ErrorNoValidImage = 0 (if an image is present) ErrorInvalidVersion = any valid value Payload = [0x]05XX	The firmware upgrade image must be deleted from the DUT if available.

1159

1160 7. Multi-Package Access

1161 Test Purpose: This test verifies all the Multi-Package Access commands from TS007-
1162 Multi_Package_Access-v1.0.0 [7].

1163

- 1164 **Pre-Requisite for Multi Package Access protocol**
- 1165 DUT clock synchronization:
- 1166 • TCL sends CP-CMD DeviceTimeReq on FPort 224 to DUT.
 - 1167 • DUT must issue a DeviceTimeReq command.
 - 1168
- 1169 **PackageVersion**
- 1170 • TCL sends the command PackageVersionReq to the DUT on FPort 225
 - 1171 • DUT must respond with the command PackageVersionAns with Package Identifier = 0 and
 - 1172 Package Version = 1
- 1173 **DevPackageReq**
- 1174 • TCL sends the command DevPackageReq to the DUT on FPort 225
 - 1175 • DUT must respond with the command DevPackageAns
- 1176 Note: Answer will depend on supported packages and each supported package FPort
- 1177 number
- 1178
- 1179 **Multi-Package Access Protocol**
- 1180 • If the DUT supports any of the packages, TCL sends multiple commands with FPort = 225
 - 1181 for each of these packages with
 - 1182 PackageIdentifier
 - 1183 CMD
 - 1184 Payload
 - 1185 ...
 - 1186 Token = [0x]00
 - 1187 • DUT must respond to each of the commands supported
 - 1188 PackageIdentifier
 - 1189 CMD
 - 1190 Payload
 - 1191 ...
 - 1192 Token = [0x]00
 - 1193 • If multicast and Class B/Class C is supported,
 - 1194 ○ Wait for Session time
 - 1195 ○ TCL sends Unconfirmed frame using multicast and session context
 - 1196 ○ DUT must silently drop the multicast downlink
 - 1197 ○ Wait for Session time out
 - 1198
 - 1199 • TCL switches to Lowest DR, refer [2]
 - 1200 • TCL sends multiple commands using FPort = 225

- 1201 CMD PackageVersionReq
1202 Payload
1203 (Repeat 'X' times until Payload response will be > Max Payload [2])
1204 Token = [0x]01
- 1205 • DUT must respond with CMD MultiPackBufferFrag using FPort = 225 since response
1206 payload > max payload
1207 CMD MultiPackBufferFrag
1208 BaseByte
1209 ANS Buffer
1210 CMD PackageVersionAns
1211 Payload
1212 (Repeat 'X' times until Payload response (ANS buffer length + 1) is equal to the
1213 maximum payload length authorized for the lowest DR, refer [2])
1214 Payload
1215 Token = [0x]01
 - 1216 • DUT must repeat the CMD MutliPackBufferFrag for the remaining buffer content
1217
 - 1218 • TCL sends multiple commands using FPort = 225
1219 • DUT must respond with CMD MultiPackBufferFrag using FPort = 225 since response
1220 payload > max payload
1221 • While the buffer is being transmitted, TCL sends CMD MultiPackBufferFragReq
1222 • DUT must abort the transmission of the buffer and transmit the newly requested buffer
1223 segment
1224
 - 1225 • TCL sends multiple commands using FPort = 225
1226 • DUT must respond with CMD MultiPackBufferFrag using FPort = 225 since response
1227 payload > max payload
1228 • While the buffer is being transmitted, TCL sends CMD PackageVersionReq
1229 • DUT must abort the transmission of the buffer and respond to the new request
1230
 - 1231 • TCL sends the command MultiPackBufferReq to the DUT using FPort 225 – with an invalid
1232 StartByte and StopByte
1233 • DUT must respond with MultiPackBufferFrag with Error = [0x]FF
1234 • TCL sends the command MultiPackBufferReq to the DUT using FPort 225 – where
1235 StopByte is lesser than StartByte
1236 • DUT must respond with MultiPackBufferFrag with Error = [0x]FF
1237 • TCL sends the command MultiPackBufferReq along with another command
1238 • DUT must silently discard the downlink
1239
 - 1240 • TCL sends the command MultiPackBufferReq to the DUT using FPort 225 – with a valid
1241 StartByte and StopByte such that the requested bytes are greater than the maximum
1242 payload length.
1243 • DUT must send with the command MultiPackBufferFrag containing the

- 1244 ○ BaseByte field – encodes the index of the first byte of the buffer being
1245 transmitted. The BaseByte field is always ≤ 127
1246 ○ ANS buffer - It contains all the requested bytes of the ANS buffer starting from
1247 ByteBase byte, until the maximum payload length is reached
1248 • When the maximum payload length is reached, another fragment must be sent by the
1249 DUT to complete the transmission of the ANS buffer.
1250
1251 • TCL sends a LinkADRReq to change the DataRate to Max125kHzDR. DUT must send a
1252 successful LinkADRAns.
1253 • If a multicast group was created, TCL sends a McGroupDeleteReq to delete the McGroup
1254 created. DUT must respond successfully with a McGroupDeleteAns.
1255
1256
1257

1258 Test Procedure Frame Sequence Chart

1259

Step	Procedure	Frame Sequence		Test Purpose
		End Device - TCL	Frame	
1	DUT sends Unconfirmed frame	→		
	TCL sends Unconfirmed frame	←	CP-CMD DeviceTimeReq FPort = 224 Payload = [0x]21	For device clock sync
2	DUT sends Unconfirmed frame	→	MAC-CMD DeviceTimeReq FPort = 0 Payload = [0x]0D	
	The TCL sends Unconfirmed frame	←	MAC-CMD DeviceTimeAns FPort = 0 Payload = [0x]0DXXXXXXXXXX	
3	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD PackageVersionReq FPort = 225 Payload = [0x]00 Token = [0x]00	
4	DUT sends Unconfirmed frame	→	CMD PackageVersionAns FPort = 225 Package Identifier = 0 Package Version = 1 Payload = [0x]000001 Token = [0x]00	
5	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD DevPackageReq FPort = 225 Payload = [0x]01 Token = [0x]01	
6	DUT sends Unconfirmed frame	→	CMD DevPackageAns FPort = 225 NbPackages = number of packages implemented ... Package Identifier A Package Version A Port A ... (for each package implemented) Port Payload = [0x]01XX Repeat XXXXXX for each package implemented Token = [0x]01	

	TCL sends Unconfirmed frame	←	FPort = 225 <i>[If DUT supports Clock Sync]</i> PackagIdentifier = 1 CMD PackageVersionReq Payload = [0x]00 <i>[If DUT supports Multicast]</i> PackagIdentifier = 2 CMD McGroupSetupReq Payload = [0x]02 XX(for 29 octets) <i>[If DUT supports Class B]</i> CMD McClassBSessionReq Payload = [0x]05 XX(for 10 octets) <i>[If DUT does not support Class B and supports Class C]</i> CMD McClassCSessionReq Payload = [0x]04 XX(for 10 octets) Note: The parameters used for these commands will be the same as Section 4 Remote Multicast Setup. <i>[If DUT supports Fragmentation]</i> PackagIdentifier = 3 CMD PackageVersionReq Payload = [0x]00 <i>[If DUT supports Firmware Upgrade]</i> PackagIdentifier = 4 CMD PackageVersionReq Payload = [0x]00 PackagIdentifier = 0 CMD PackageVersionReq Payload = [0x]00 Token = [0x]02	Valid field values for each command must be set as described in sections above
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7	DUT sends Unconfirmed frame	→	FPort = 225 <i>[If DUT supports Clock Sync]</i> PackagelIdentifier = 1 CMD PackageVersionAns Payload = [0x]000102 <i>[If DUT supports Multicast]</i> PackagelIdentifier = 2 CMD McGroupSetupAns Payload = [0x]0200 <i>[If DUT supports Class B]</i> CMD McClassBSessionAns Payload = [0x]05XXXXXXXX <i>[If DUT does not support Class B and supports Class C]</i> CMD McClassCSessionAns Payload = [0x]04XXXXXXXX <i>[If DUT supports Fragmentation]</i> PackagelIdentifier = 3 CMD PackageVersionAns Payload = [0x]000302 <i>[If DUT supports Firmware upgrade]</i> PackagelIdentifier = 4 CMD PackageVersionAns Payload = [0x]000401 PackagelIdentifier = 0 CMD PackageVersionAns Payload = [0x]000001 Token = [0x]02	
	If DUT supports Multicast and Class B/Class C, perform Steps 3 and 4			
8	After Session Time, TCL sends Unconfirmed frame using multicast and session context	←	FPort = 225 CMD PackageVersionReq Payload = [0x]00 Token = [0x]03	
9	DUT sends Unconfirmed frame	→		DUT silently drops the downlink
	Wait Until Session Time Out			
10	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CP-CMD RegionalDutyCycleCtrlReq-OFF Payload = [0x]0500 FPort = 224	Turn OFF Regional Duty Cycle

11	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CMD LinkADRRReq DataRate = lowest DR, refer [2] Payload = [0x]03XXXXXXXXXX	Switch to lowest DR
12	DUT sends Unconfirmed frame	→	CMD LinkADRAns Payload = [0x]0307	
	The TCL sends Unconfirmed frame	←	FPort = 225 CMD PackageVersionReq Payload = [0x]00 (Repeat 'X' times until Payload response will be > Max Payload [2]) Token = [0x]03	
13	DUT sends Unconfirmed frame	→	FPort = 225 CMD MultiPackBufferFrag BaseByte = index of 1 st byte of the buffer transmitted (<=127) ANS buffer – CMD PackageVersionAns Package Identifier = 0 Package Version = 1 Payload = [0x]00XXXXX (Repeat 'X' times until Payload response (ANS buffer length + 1) is equal to the maximum payload length authorized for the lowest DR, refer [2]) Payload = [0x]02XXXXX (var)XX 1-MaxPayloadLen-3 Token = [0x]03	
14	DUT sends Unconfirmed frame	→	FPort = 225 CMD MultiPackBufferFrag BaseByte = index of continuation byte of the buffer transmitted (<=127) ANS buffer – [CMD PackageVersionReq Package Identifier = 0 Package Version = 1 Payload = [0x]00XXXXX ...] (Repeat 'X' times until end of response) Payload = [0x]02XXXXX (var)XX 1-MaxPayloadLen-3 Token = [0x]03	Completes transmission of ANS buffer

	The TCL sends Unconfirmed frame	←	FPort = 225 CMD PackageVersionReq Payload = [0x]00 (Repeat 'X' times until Payload response will be > Max Payload [2]) Token = [0x]00	
15	DUT sends Unconfirmed frame	→	FPort = 225 CMD MultiPackBufferFrag BaseByte = index of 1 st byte of the buffer transmitted (<=127) ANS buffer – CMD PackageVersionAns Package Identifier = 0 Package Version = 1 Payload = [0x]00XXXX (Repeat 'X' times until Payload response (ANS buffer length + 1) is equal to the maximum payload length authorized for the lowest DR, refer [2]) Payload = [0x]02XXXX (var)XX 1-MaxPayloadLen-3 Token = [0x]00	
	While the buffer is being transmitted, the TCL sends Unconfirmed frame	←	CMD MultiPackBufferReq FPort = 225 StartByte = start byte present in the ANS buffer already transmitted StopByte – stop byte present in the ANS buffer already transmitted Payload = [0x]02XXXXXXXX	
16	DUT sends Unconfirmed frame DUT aborts transmission and transmits the newly requested ANS buffer segment	→	FPort = 225 CMD MultiPackBufferFrag BaseByte = start byte requested ANS buffer – contains all bytes requested in the request Payload = [0x]02XXXX (var)XX Token = [0x]00	DUT aborts transmission and transmits the newly requested ANS buffer segment
	The TCL sends Unconfirmed frame	←	CMD MultiPackBufferReq FPort = 225 StartByte and StopByte – Invalid set of bytes Payload = [0x]02XXXXXXXX	
17	DUT sends Unconfirmed frame	→	CMD MultiPackBufferFrag Error = 0xFF Payload = [0x]02XX Token = [0x]00	
	The TCL sends Unconfirmed frame	←	CMD MultiPackBufferReq FPort = 225 StopByte is less than StartByte Payload = [0x]02XXXXXXXX	

18	DUT sends Unconfirmed frame	→	CMD MultiPackBufferFrag Error = 0xFF Payload = [0x]02XX Token = [0x]00	
	The TCL sends Unconfirmed frame	←	FPort = 225 CMD MultiPackBufferReq FPort = 225 StartByte = start byte present in the ANS buffer already transmitted StopByte – stop byte present in the ANS buffer already transmitted Payload = [0x]02XXXXXXXXX CMD PackageVersionReq Payload = [0x]00 Token = [0x]01	
19	DUT sends Unconfirmed frame	→		DUT must silently discard the downlink
	The TCL sends Unconfirmed frame	←	FPort = 225 CMD PackageVersionReq Payload = [0x]00 (Repeat 'X' times until Payload response will be > Max Payload [2]) Token = [0x]01	
20	DUT sends Unconfirmed frame	→	FPort = 225 CMD MultiPackBufferFrag BaseByte = index of 1 st byte of the buffer transmitted (<=127) ANS buffer – CMD PackageVersionAns Package Identifier = 0 Package Version = 1 Payload = [0x]00XXXX (Repeat 'X' times until Payload response (ANS buffer length + 1) is equal to the maximum payload length authorized for the lowest DR, refer [2]) Payload = [0x]02XXXX (var)XX 1-MaxPayloadLen-3 Token = [0x]01	
	While the buffer is being transmitted, the TCL sends Unconfirmed frame	←	FPort = 225 CMD PackageVersionReq Payload = [0x]00 Token = [0x]02	

21	DUT sends Unconfirmed frame	→	FPort = 225 CMD PackageVersionAns Package Identifier = 0 Package Version = 1 Payload = [0x]00XXXX Token = [0x]02	DUT aborts transmission and responds to the new request
	The TCL sends Unconfirmed frame	←	MAC-CMD LinkADRReq DataRate = Max125kHzDR, refer [2] Payload = [0x]03XXXXXXXXXX	
22	DUT sends Unconfirmed frame	→	MAC-CMD LinkADRAns Payload = [0x]0307	
	If a multicast group was created, the TCL sends Unconfirmed frame	←	CMD McGroupDeleteReq McGroupID = 0 Payload = [0x]03XX	
23	DUT sends Unconfirmed frame	→	CMD McGroupDeleteAns McGroupUndefined = 0 McGroupID = 0 Payload = [0x]03XX	
24	DUT sends Unconfirmed frame	→		
	The TCL sends Unconfirmed frame	←	CP-CMD RegionalDutyCycleCtrlReq-ON Payload = [0x]0501 FPort = 224	Turn ON Regional Duty Cycle

1260